Does Easing Migration Barriers Increase Labor Unrest? Evidence from the Hukou Reform in China*

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Abstract

A large body of evidence suggests that granting formal status to migrants reduces crime and unrest. Our results challenge this view. China's 2014 Hukou reform eased within-country migration barriers, most extensively for cities below one million residents. Exploiting this variation, we estimate the reform increased per capita unrest by about 2.5 times its pre-reform average. The increase was concentrated in manufacturing, construction, and transportation, sectors with high migrant employment. Manufacturing employment did not increase, and earnings declined in low-skill services. We argue that formal status expanded migrants' local social and economic entitlements faster than employers and local governments fulfilled them.

Keywords: Hukou Reform; Internal Migration; Labor Unrest; Urban Labor Markets
JEL Codes: J52, J61, J68, O15, R23

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1 Introduction

Reducing migration barriers is among the most promising policy levers for economic growth,[1] yet the social consequences of such reforms are far less understood. Does easing migration barriers reduce labor unrest? Most evidence on international migrants suggests yes. Granting formal legal status reduces crime and social conflict, with causal evidence from high-income settings where formal employment absorbs new workers at stable wages (Marie and Pinotti, 2024). Whether this pattern extends to internal migration in developing countries, where labor markets face distinct structural constraints, including the persistence of surplus labor, remains an open question. In this paper, we examine this question using China's 2014 Hukou reform and find evidence that runs counter to the conventional wisdom. In fact, relative to control cities, per capita unrest in the cities that received the most extensive barrier reduction rose by 0.104 events per 100,000 residents, about 2.5 times the pre-reform mean of 0.041.

China's Hukou system restricts more than 145 million rural workers from full access to public services in their destination cities (Gao et al., 2023). By conditioning access to education, healthcare, and social welfare on a worker's place of birth, the system creates a large population of urban residents without formal residency rights. The 2014 reform relaxed these requirements based on city size, assigning the most extensive barrier reduction to cities below one million residents and only partial reduction to cities between one and five million. The gradient in reform intensity provides the identifying variation for this paper.

The population thresholds were set by the central government as part of a national urbanization strategy, not in response to local unrest trajectories. Pre-reform trends in treatment and control cities track each other closely across seven years. Exploiting this variation in a difference-in-differences design, we find that per capita unrest in treatment cities rose sharply after 2014, with the differential building through 2016 before attenuating. The registered population grew 6–11 percent faster in treatment cities over the same period. The sectoral pattern provides the first piece of mechanism evidence. A mechanism operating through migrant labor markets predicts unrest in the sectors migrants can enter and none in the sectors they cannot. That is what we find. The increase concentrates in manufacturing, construction, and transportation, where migrants most often seek work, while credentialed services show no detectable change.

[1] Empirical estimates suggest that eliminating international migration barriers could raise global GDP by 50–150 percent (Clemens, 2011), while easing internal barriers can increase labor productivity by more than 20 percent (Bryan and Morten, 2019).

The key insight is that the reform changed workers' formal status faster than their economic conditions. Formal residency went to migrants already working in treatment cities and became easier to obtain for new arrivals, and on both margins it carried entitlements to local social insurance and public services and a permanent stake in the destination city, raising the conditions workers expected from employers and local governments. The registered population rose 6–11 percent while the physical population showed no detectable change (Lai and Qiu, 2025), so the dominant margin was formalization of workers already present. The expected conditions did not materialize. Manufacturing showed no differential growth after the reform, service employment expanded only in low-skill sectors such as transportation, where piece-rate earnings fell as more workers entered,[2] and wage arrears remained the dominant grievance, cited in 74 percent of unrest events with identified causes. We identify three reinforcing channels as part of our mechanism analysis. Formalization widened the gap between what workers were promised and what they received, congestion in the few sectors migrants can enter rationed jobs and compressed earnings, and permanent residency lowered the cost of acting collectively on those grievances.[3] The mechanism section evaluates this explanation using evidence on employment, wages, and fiscal capacity, and assesses its plausibility relative to leading alternative accounts. Explanations based on native backlash or congestion in public services require a physical inflow of migrants that did not occur. Likewise, changes in incident reporting cannot readily explain increases concentrated in migrant-intensive sectors while incidents in the service sector remain unchanged.

These findings suggest that the sectoral composition of local labor markets, rather than aggregate economic growth alone, shapes whether migration reform promotes stability or generates unrest. When formal status is granted before migrants are fully integrated into the labor market and labor demand in accessible sectors adjusts slowly, migration reform can increase instability even amid sustained aggregate growth. An et al. (2024) document the migrant wage compression that the same reform created. We document the city-level collective consequence of that compression, showing that unrest rose where formalized workers entered sectors that did not expand. A related concern may arise in other household-registration settings, including Vietnam, where formal registration status similarly determines access to local services (Huỳnh, 2025).

[2] Piece-rate earnings in sectors such as transportation refer to the contractual practices where workers (i.e., drivers) are paid per trip or per delivery.

[3] At the time of writing this paper, supporters of an anti-migrant referendum in Switzerland aimed to cap the total population of the country, claiming that the influx of migrant workers "puts housing, schools, transport, welfare and the Swiss way of life itself under unbearable strain" (Henley, 2026).

Related Literature. A large body of research has documented significant internal migration costs that limit workers' ability to realize productivity and welfare gains from moving, including institutional barriers that restrict Hukou-registered workers in China (Ngai et al., 2019; Fan, 2019; Hsu and Ma, 2021; Wang et al., 2021; Jin and Zhang, 2023) and spatial frictions that reduce labor mobility more broadly (Imbert and Papp, 2020; Bryan and Morten, 2019). At the firm level, Imbert et al. (2022) show that large Chinese firms absorb internal migrants through expansion rather than wage adjustment, a channel that may not operate in the smaller cities where the 2014 reform had its deepest effect. Whether reducing these frictions affects social stability in destination cities is a different and less studied question. Most existing evidence on migration and conflict comes from contexts where migration is driven by displacement or economic shocks, in which observed social tensions may reflect the forced nature of inflows rather than the effects of policy itself. Even when governments deliberately ease internal migration barriers, reforms either apply uniformly across regions, leaving no variation to identify causal effects, or operate through infrastructure investment that simultaneously changes trade costs and market access, confounding the migration channel.

The 2014 Hukou reform in China is an exception. It changed administrative residency requirements without altering physical infrastructure or trade costs, and created a gradient of reform intensity across cities with different population sizes. Previous research using natural experiments finds that granting formal legal status to international migrants reduces individual crime (Pinotti, 2017; Mastrobuoni and Pinotti, 2015; Bell et al., 2013; Marie and Pinotti, 2024). That mechanism is employment-based. Legal status removes a ban on formal work and raises returns to legitimate activity. In Switzerland, the removal of immigration restrictions increased firm performance and worker productivity as labor markets absorbed inflows through firm creation and wage adjustment (Beerli et al., 2021). The Hukou system operates differently. Migrants participated in local labor markets regardless of their Hukou status. Gaining formal residency changed what workers could claim in the city, local public services, social insurance, and a permanent right to stay, rather than whether they could work, and the outcome of interest is collective labor action rather than individual crime. The two sets of findings need not conflict. Where labor markets are tight, employers have to honor what formal status promises in order to keep their workers, so conditions improve with status and conflict falls. The legalizations studied by Pinotti (2017) and Marie and Pinotti (2024) took place in such markets. Where labor is abundant relative to the jobs migrants can reach, employers face no pressure to honor the new status. Pay falls where it is flexible and queues form where it is not, so the gap between promised and actual conditions widens and conflict rises. The Hukou reform took place in the second kind of labor market. An et al. (2024) show that migrant wages fell by 2.6–7.9 percent on average in non-megacities after the reform, with the declines growing monotonically as city size falls. We document the city-level unrest that followed.

Although granting legal status reduces individual crime, a separate literature finds that migration inflows can increase collective conflict when receiving areas lack absorptive capacity. Helms (2024) finds that a 10 percentage point increase in migrant inflows, induced by India's trade exposure, increased the incidence of anti-migrant riots by 6.5 percentage points. Knight and Tribin (2023) show that Venezuelan displacement to Colombian border municipalities increased homicide rates, driven by the victimization of migrants rather than crimes they committed. McGuirk and Nunn (2024) provide the closest parallel to the mismatch mechanism we document. They show that agricultural development projects implemented in traditionally pastoral regions of Africa nearly doubled the risk of conflict when local economies were unable to absorb the resulting disruption. All three papers, along with ours, share a common underlying logic: conflict arises when labor inflows or policy interventions—and the speed at which they occur—outpace the capacity of receiving environments to absorb them.

Our paper differs from these studies primarily in the type of migration analyzed and the type of conflict studied. The conflict these studies document reflects anti-migrant backlash, physical vulnerability, or resource competition from migration driven by forces outside policy control. The unrest we document is collective labor conflict concentrated in manufacturing, construction, and transportation sectors, arising from a targeted government policy that expanded migrants' access to formal residency in these cities. Within the Chinese context, prior work examines the structural conditions and state responses that shape collective unrest (Elfstrom and Kuruvilla, 2014; Cai and Chen, 2022; Lorentzen et al., 2013; Qin et al., 2024; Wen, 2025), but no study identifies a specific policy reform as a causal driver.

We are not the first to study this question in the Chinese context. The closest to our study is Lai and Qiu (2025), who examine the 2014 Hukou reform and report lower unrest using a regression discontinuity design at the 3 million population threshold. However, our work differs from theirs in three fundamental ways, which also help with our identification. First, we estimate the average treatment effect across cities where the reform eliminated or substantially reduced migration requirements, not a local effect at the boundary where the contrast is between partial reform and no reform. Second, we exploit a five-group dose-response gradient that reveals monotone treatment heterogeneity pooled away by a single treatment indicator. Third, our panel provides six pre-reform years of parallel trends evidence. Section 3.2 examines these contributions in detail. On the policy's other margin, Guo et al. (2025) use linked employer-employee data to study a city-level tightening of employer-sponsored Hukou quotas and find that scarcer quotas raised migrants' relative wages, because employers could extract less of a compensating differential for sponsoring residency, while pushing skilled migrants toward quota-rich public-sector employers. Their results are consistent with a central premise of our argument—that Hukou status is valuable enough to be directly embedded in employment relationships—while our evidence instead focuses on its implications for collective action across cities.

The remainder of the paper is organized as follows. Section 2 presents the research design and data. Section 3 reports the main results. Section 4 investigates mechanisms. Section 5 concludes.

2 Research Design

The central challenge in identifying how easing migration barriers affects labor unrest is separating the reform's direct impact from other forces that shape city-level labor conditions over the same period. The 2014 reform's city-size classification provides a credible source of identification because the population thresholds determining reform intensity were set centrally as part of a national urbanization strategy, announced only in mid-2014, and bear no direct connection to pre-existing unrest patterns at the city level.

2.1 The 2014 Hukou Reform

In July 2014, the Chinese government announced a landmark reform of the household registration system (Hukou) that had shaped China's social and economic structure for over half a century. Originally introduced in the 1950s, the Hukou system tied individuals to their place of birth, classifying them as rural or urban residents and determining their access to education, healthcare, housing, and social welfare. The system created stark divisions between rural and urban populations while limiting the mobility of hundreds of millions of rural migrants seeking better opportunities in cities.

The 2014 reform, implemented through the "National New-Type Urbanization Plan (2014–2020)" and related State Council directives, aimed to encourage rural migrants to gain formal residency status in urban areas as part of China's broader urbanization strategy. The reform expanded the eligibility of migrants for local public services, including education, social security, healthcare, and public housing (Zhang and Treiman, 2013; Wu and Treiman, 2004), narrowing the welfare gap between migrants and registered urban residents (Xie and Zhou, 2014).

The 2014 reform classified Chinese cities into five categories based on the size of their urban population, each implementing differentiated approaches to reducing Hukou requirements (An et al., 2024). Cities with populations less than 0.5 million were required to remove all prior administrative restrictions, with stable formal employment as the sole remaining pathway to local Hukou. Medium-sized cities (0.5–1 million) experienced substantial reduction in requirements, with migrants needing one to three years of social security contributions along with stable employment. Larger cities (1–3 million residents) maintained controlled issuance, requiring three to five years of contributions, while cities with 3–5 million residents imposed even stricter criteria. Megacities such as Beijing and Shanghai, with populations exceeding 5 million, maintained strict controls with no significant reduction in requirements. In our empirical analysis, we map these five categories to a binary assignment. Cities in the two smallest categories (populations below 1 million, sizes 4 and 5) form the treatment group, which received the most extensive reduction in Hukou requirements. Cities in the two intermediate categories (populations between 1 and 5 million, sizes 2 and 3) form the control group, which experienced only partial barrier reduction. We use "treatment" and "control" throughout.

The Hukou registration confers permanent formal residency in the destination city, entitling holders to local public services, education, and social insurance. The reform therefore targeted a barrier different from the temporary residence permits that govern short-term labor mobility. The population response we document reflects permanent or long-term in-migration rather than circular flows.

2.2 Measurement of Labor Unrest

We obtained the data on labor unrest, including labor strikes and protests, from the China Labor Bulletin (CLB; China Labor Bulletin, 2024), a Hong Kong-based non-governmental organization dedicated to supporting the development of trade unions within China. This dataset is widely used in research on labor unrest, labor rights, and state-society relations (Qin et al., 2024) and is considered one of the most comprehensive sources of information on strike and protest events in China. Since 2007, the CLB has systematically collected labor strike and social protest statistics, with electronic coverage available from 2011 to date.[4] The CLB data measure reported incidents of labor unrest, including labor strikes,

[4]For strike data prior to 2011, similar information was extracted from annual CLB reports and supplemented with additional sources such as Boxun, a U.S.-based political news website targeting Chinese audiences. Coverage of the smallest cities was particularly sparse before 2011, with only 3–5 total events per year across all 51 size-5 cities in 2007–2010.

protests, demonstrations, and other forms of collective action. Each record identifies the timing, prefecture-level city location, employer name and description, industry classification, number of participants, worker actions, and government responses. The CLB data compilation draws from multiple sources, including overseas Chinese media, labor activists within China, and internet searches that cover social media platforms.

Our analysis examines labor unrest events from 2007 to 2019, a period longer than comparable studies in this literature. The sample begins in 2007, when the CLB commenced systematic data collection, and ends in 2019 to avoid confounding from COVID-19 lockdowns and concurrent changes in internet censorship enforcement that disrupted both labor market conditions and the CLB's collection methods after 2019.[5] The CLB recorded approximately 12,700 unrest events during this time span. We split events by the recorded worker action. Actions whose description contains "strike" are coded as strikes, and the remaining collective actions, including demonstrations, sit-ins, road blockages, petitions, demands for government intervention, and threats to jump, are coded as protests. Under this coding, protests form the majority, approximately 9,400 events against approximately 3,200 strikes. Approximately 78 percent of incidents involve fewer than 100 participants, and wage arrears are the primary cause, cited in approximately 74 percent of events with identified grievances.

In addition to the CLB, we rely on city-level data from the China Statistical Yearbook (National Bureau of Statistics of China, 2019). We use 2013 urban population to classify cities into size categories, and annual registered population for per capita rates, employment, and fiscal variables for the mechanism analysis.[6] Our final dataset includes 285 prefecture-level cities, representing around 85 percent of all Chinese prefecture-level cities, and providing coverage comparable to most existing studies examining the 2014 Hukou reform. Throughout the paper, "city" refers to a prefecture-level city, China's principal subnational administrative unit below the province. All fixed effects, clustering, and treatment assignments operate at this level. The sample contains 285 cities distributed across the five size categories, with 12 megacities (Category 1), 11 large cities (Category 2), 108 medium cities (Category 3), 103 small cities (Category 4), and 51 towns (Category 5).

We construct three outcome variables from these events. *Labor Unrest* is the total count of incidents in a city in a given year t, while *Strike* and *Protest* count the two categories

[5]CLB coverage for 2009 is substantially below adjacent years (35 total incidents versus 69–90 in 2007–2008 and 2010, Appendix Table A1), but excluding it does not materially affect the estimates.

[6]We use only consistent population measures. In particular, remote or smaller cities with missing urban population data are excluded from the analysis due to incomplete observations. These cities together represent only a minor share of national GDP and economic activity.

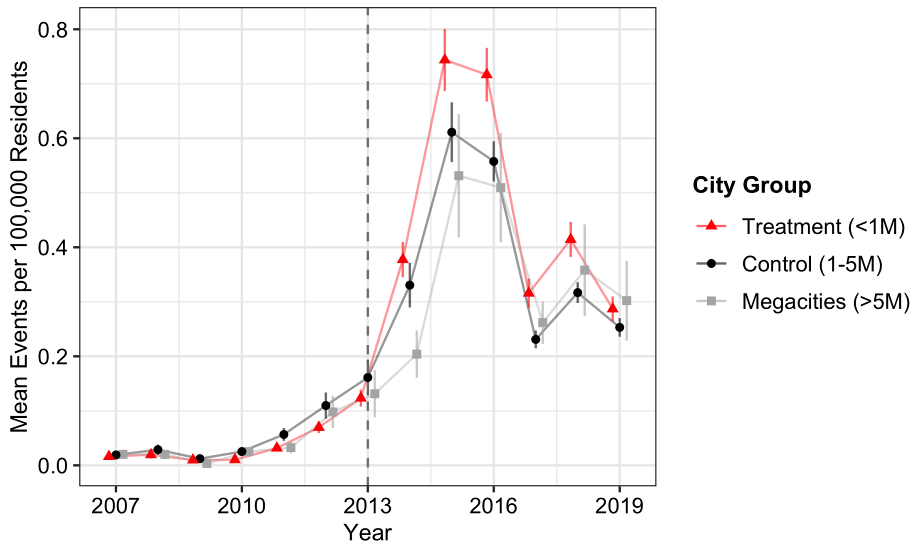
defined above. This disaggregation allows us to distinguish work-stoppage actions from other forms of collective labor grievance, which may respond differently to the Hukou reform.

For each of these three outcomes, we examine two complementary measures, the raw event count and the per capita rate (events per 100,000 registered population). The appropriate measure of labor unrest exposure is incidents per resident, not total incident volume. A city of five million with ten strikes and a city of 500,000 with ten strikes present identical events to a count-based measure, yet residents of the smaller city experience ten times the disruption per capita. Per capita rates are particularly relevant here because the reform itself induced differential population growth across city groups. When differential population growth alters the denominator, raw counts that appear comparable across groups can mask divergent per capita rates. Per capita unrest rose because incidents grew faster than the expanding registered population, a divergence our raw counts understate. Therefore, we treat per capita rates as the primary outcome variable. Per capita unrest rates are right-skewed with many zero city-year observations. OLS imposes a linear conditional mean on a non-negative outcome with this shape. Section 3.4 confirms that PPML, which requires no linearity assumption for non-negative outcomes, produces estimates that agree with OLS in direction and significance.

Two features of CLB coverage need further discussion. First, event reporting expanded substantially over the sample period. Appendix Table A1 documents total incident counts by year and city size. The CLB recorded fewer than 100 events per year before 2011, rising to over 600 in 2013 and peaking at roughly 2,600 in 2015–2016 before declining to about 1,300 by 2019. This growth reflects the CLB's shift to electronic crowd-sourced collection after 2011 and the broader expansion of social media reporting in China, not necessarily a proportional increase in actual unrest. Second, the reporting growth was not uniform across city sizes. Smaller cities, which had the sparsest coverage in 2007–2010 (as few as 3–5 events per year across all 51 size-5 cities), gained disproportionately from electronic coverage expansion. If reporting improvements were concentrated in treatment cities, the post-reform increase could partly reflect better detection rather than more unrest. Section 3.4 examines this concern in detail.

Figure 1 displays the mean per capita labor unrest from 2007–2019 for treatment cities, control cities, and megacities. All three groups have similar rates before 2014 and all three rise sharply after 2013, with treatment cities showing the largest increase. Both the common increase and the differential are visible in the raw data, and the difference-in-differences design isolates the differential.

Figure 1: Mean Per Capita Labor Unrest by City Group (2007–2019)



Notes: The figure plots mean per capita labor unrest (strikes and protests combined, events per 100,000 registered population) by year for treatment cities (population below 1 million, sizes 4–5), control cities (population 1–5 million, sizes 2–3), and megacities (population above 5 million, size 1). The dashed vertical line marks 2013, the last pre-reform year.

2.3 Identification Strategy

We adopted a difference-in-differences event study design to estimate how variation in the intensity of Hukou barrier reduction across city types affected labor unrest. The design compares cities that received the most extensive reduction in migration barriers (the treatment group) to cities where restrictions were only modestly eased (the control group). Treatment cities are those with 2013 urban populations below 1 million (sizes 4 and 5 in the reform's classification scheme), where the 2014 policy mandated complete or near-complete elimination of Hukou requirements. Control cities have populations between 1 and 5 million (sizes 2 and 3), which experienced only partial barrier reduction through requirements such as multi-year social security contributions or stable employment thresholds.

We used 2013 urban population for classification to avoid potential endogeneity from cities adjusting reported population in response to the reform. Since the specific population thresholds were formalized only in mid-2014, the pre-announcement population reflects predetermined city characteristics rather than strategic responses to the policy.[7]

[7] As a robustness check, we verify that results are quantitatively similar when using 2014 population for classification (Appendix Table A3). Only 17 of 285 cities (6 percent) change size category between 2013 and 2014, confirming that the classification is stable.

The treatment group includes 154 cities and the control group includes 119 cities, yielding about 3,500 city-year observations over the 2007–2019 panel. Appendix Table A2 reports pre-reform sample characteristics. Treatment cities are smaller by construction, since treatment is assigned by city size, and have correspondingly lower population, employment, and fiscal levels. Pre-reform per capita unrest rates are modestly lower in treatment cities (0.04 versus 0.06 per 100,000). These level differences are absorbed by city fixed effects and do not threaten identification. The key diagnostic is the parallel pre-reform trends assumption, which we examine in Section 3.

We conduct the baseline comparison between the full and partial migration barrier reduction cities, and set aside megacities (populations above 5 million, size 1) for now. The reasoning is about comparability. Comparing the two reform arms holds the policy environment closer to constant and isolates the gradient in reform intensity. Megacities received no barrier reduction, differ from small and medium cities in political status and economic structure, and the sample contains only 12 of them. Their financial-center economies were also exposed to the mid-2015 stock market crash and the intensified 2015 anti-corruption campaign in ways smaller cities were not, although the post-2013 rise in unrest itself was common to all city groups. The raw means in Figure 1 peak in 2015 at 0.74 events per 100,000 in treatment cities, 0.61 in control cities, and 0.53 in megacities, so identification does not rest on megacities being uniquely shocked, and the design nets out the common rise. Section 3.4 shows that the strike results are robust to using megacities as the comparison group.

The binary contrast is also a deliberate estimation choice. The reform assigns five intensity levels, and one could instead regress unrest on a multi-valued or continuous treatment. With treatment effects that differ across intensity levels, however, a single coefficient on such a treatment aggregates the level-specific effects with weights that can be negative and need not answer any policy question (Callaway et al., 2024). The binary comparison of the most against the least treated identifies a transparent average effect under standard parallel trends. The cost is pooling across intensity levels, which we address by reporting the full five-group dose response in Section 3.

To examine our research question, we estimate the following event study specification for the period 2007–2019:

$$y_{it} = \sum_{k=-6, k\neq 0}^{6} \beta_k \times \text{Treat}_i \times \mathbf{1}(t = 2013 + k) + \alpha_i + \gamma_{p(i)t} + \varepsilon_{it} \tag{1}$$

where y_{it} represents the outcome of interest in city i and year t, so each observation is at the city-year level. The treatment indicator, Treat_i, equals one for treatment cities

(2013 population below 1 million) and zero for control cities (population between 1 and 5 million). The event-time coefficients β_k capture the differential change in outcomes for treatment cities relative to control cities in year $2013 + k$ compared to the omitted reference year 2013, with k ranging from -6 (year 2007) to 6 (year 2019). City fixed effects α_i absorb time-invariant city characteristics, while $\gamma_{p(i)t}$ denotes province-by-year fixed effects for province $p(i)$ of city i, controlling for all time-varying shocks at the province level, including common trends, provincial economic conditions, and concurrent reforms. Standard errors are clustered at the city level to account for arbitrary serial correlation within cities.

Our estimation design identifies the average treatment effect of reform intensity on city-level per capita unrest, treating the 1 million population threshold as the assignment rule to the treatment group. The dependent variable is defined as unrest events per 100,000 registered (Hukou) population. Because the registered population itself is an administrative outcome of the reform, the rate measures unrest per 'legal' resident rather than per 'physical' resident. This measure is the policy-relevant quantity, as local governments allocate services and enforce labor standards on the basis of registered population.

The design relies on three assumptions. The first is parallel trends, which requires that treatment and control cities would have followed parallel outcome paths in the absence of the differential Hukou reform. The pre-reform event study coefficients provide a direct test of this assumption. In Appendix Table A4 we report the results of joint F-tests, which do not reject the null that pre-reform coefficients are jointly zero ($p = 0.42$ to 0.88), and we examine the pre-reform patterns for each outcome in Section 3. Flat pre-trends are necessary but not sufficient for the assumption to hold after the reform. Province-by-year fixed effects absorb all time-varying shocks common to cities within a province. However, they cannot absorb within-province shocks that load differentially on city size, as city size defines treatment assignment. In Section 3.4 we verify robustness to city-specific linear time trends, which additionally control for differential city-level trajectories.

The second assumption is no anticipation, which requires that cities did not differentially adjust their behavior before the reform took effect. The specific population thresholds defining treatment intensity were announced only through mid-2014 State Council directives (An et al., 2024), so cities had no basis to anticipate their group assignment in advance. Even if the general direction of Hukou reform was signaled at the 18th Party Congress in November 2012, responding to yet-to-be-announced numerical thresholds would have required cities to anticipate specific cutoffs that had not yet been formalized. We examine whether population trajectories in treatment and control cities diverged before 2014 in Figure 2, and also confirm the robustness of our findings to alternative city-size classification windows in Section 3.4.

A related condition requires that the population thresholds be uncorrelated with pre-existing trends in labor unrest. The thresholds correspond to China's longstanding national city-size classification system and were determined centrally as a uniform policy, not in response to local unrest trajectories. Any pre-existing level differences between city groups are absorbed by city fixed effects, and the pre-reform patterns documented in Section 3 show no evidence of differential trends before 2014.

A final concern involves spillovers across cities (SUTVA). In fact, Qin et al. (2024) document that protest in one Chinese city increases protest probability in connected cities by 17 percent within two days through social media. Because spillover protests materialize within the same calendar year as the instigating event, annual aggregation fully captures any contamination of control-city counts rather than washing it out. If treatment-city unrest spills into control cities, our DiD estimate is biased toward zero, making the positive estimates a conservative 'lower bound' on the true differential effect.

The main limitation of our model design is that the treatment assignment is determined by city size. As a result, any contemporaneous shock that differentially affected smaller cities after 2014 would confound our estimates. The province-by-year fixed effects absorb shocks common to cities within a province, including provincial policy changes, economic cycles, and enforcement campaigns. However, these fixed effects cannot absorb within-province shocks that load differentially on city size, because city size defines treatment assignment. The monotone dose-response pattern documented in Section 3, where per capita unrest increases monotonically from the largest to the smallest city category across all five size groups, strengthens the case that the Hukou reform drives our result. A confounding shock would need to generate not only a post-2014 break at the treatment threshold but also a similarly monotone pattern across the full city-size distribution. That fact narrows, but does not eliminate, the range of plausible confounders. We are not aware of other national programs that change discontinuously at the 1 million population threshold used in the Hukou reform, though we cannot rule out the possibility. Section 3.4 controls directly for the leading candidates at the national level, including i) the post-2014 trade slowdown, ii) the environmental campaigns, iii) the supply-side capacity cuts, and iv) the anti-corruption campaign. The results remain consistent after these checks. Furthermore, Appendix Figure A1 quantifies how large a violation of parallel trends would need to be to overturn the result.

We estimate Eq. (1) by OLS, but for robustness we also report the count-based Poisson pseudo-maximum likelihood (PPML) estimates in Appendix Table A5. The PPML estimates confirm our main results in direction and significance.

3 Results

We present the main findings in three steps. Section 3.1 examines differential population growth as both a validity check and the first link in the causal chain. Section 3.2 estimates the reform's effect on per capita unrest rates. Section 3.3 decomposes the result by industry sector and incident scale.

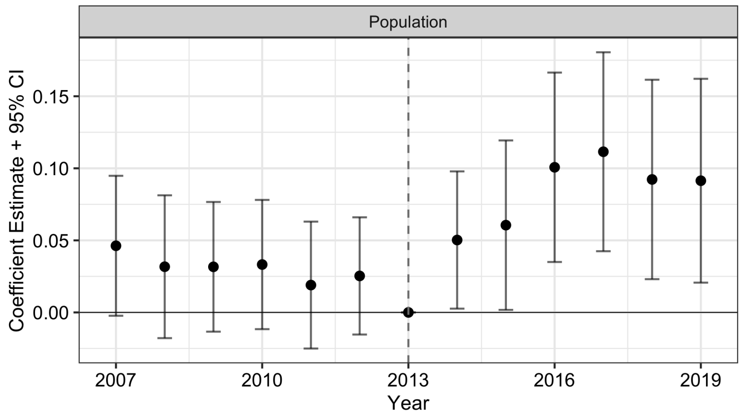
3.1 Population Response to the Hukou Reform

If the 2014 reform differentially eased migration barriers in treatment cities, the most direct observable consequence would be differential growth in the formal resident stock. We begin our analysis by examining whether this registered-population response materialized.

The registered-population response we examine measures formalization, not necessarily a net inflow of new residents, and two existing studies make the distinction concrete. Using restricted-use 2015 mini-census records, An et al. (2024) show that after 2014 migration flows tilted toward smaller administrative cities and away from larger ones, while the share of migrants who stayed within their original size group barely changed. This shift is a change in the direction of relative flows, not evidence that the total population of treatment cities rose. An et al. (2024) further caution that "getting precise measures of the true population for each city in noncensus years is always a challenge in China" (An et al., 2024, p. 175). Their data also cannot separate urban from rural destinations within a city, so part of the movement toward smaller cities may reflect migrants returning to rural hometowns rather than entering urban labor markets. Lai and Qiu (2025), examining physical urban population stocks from the Urban Construction Statistical Yearbook, find no discernible effect of the reform. When we examine registered population, the increase can reflect both the formalization of migrants already present and any new inflows the reform induced. We therefore read the registered-population response primarily as a measure of formalization in the cities where the reform bound most strongly.

Figure 2 presents event study estimates for (log) registered population, comparing the treatment and control cities. An increase in registered population suggests that more migrants gained formal residency in these cities relative to cities that retained higher Hukou requirements. The pre-reform coefficients are statistically indistinguishable from zero across all years, and a joint F-test that the 2007–2012 coefficients are zero fails to reject ($F = 1.54$, $p = 0.16$, Appendix Table A4), supporting the parallel trends assumption for population dynamics. After 2014, treatment and control cities diverge. By 2014–

Figure 2: Population Response to the Hukou Reform (2007–2019)



Notes: The figure plots event study coefficient estimates and 95% confidence intervals for log registered population, comparing treatment cities (sizes 4–5, population below 1 million) to control cities (sizes 2–3, population 1–5 million). The pooled DiD estimate is 0.058 (s.e. 0.033, $p < 0.10$, Appendix Table A6). The omitted reference year is 2013. All regressions include city and province-by-year fixed effects with standard errors clustered at the city level. Megacities (size 1, population above 5 million) are excluded.

2019, treatment cities had approximately 5–11 percent higher registered population than control cities, with the differential widening over time. Each individual post-reform year coefficient is statistically significant at the 5 percent level, though the 2014 estimate ($p = 0.04$) only narrowly clears the threshold. The pooled DiD estimate of 0.058 ($p < 0.10$, Appendix Table A6) averages across the buildup period and is therefore smaller than the peak-year coefficients.

Against a pre-reform mean of roughly 613,000 registered residents in treatment cities, this differential translates to approximately 31,000–68,000 additional Hukou holders per city.[8] We find the timing and magnitude of this population response consistent with the most extensive barrier reduction in treatment cities easing access to formal residency status, while partially reformed cities continued to restrict Hukou transfers.[9] The question that follows is whether this growth in the registered population affected labor unrest, and if so, through what channels.

[8]The pre-reform level mean of treatment-city registered population is 61.33 in units of 10,000. The event study coefficients range from 0.050 in 2014 to 0.111 in 2017–2018, corresponding to 5–11 percent growth. Appendix Table A6 reports the mean of (log) registered population (4.496), from which the level mean cannot be directly recovered due to Jensen's inequality.

[9]Splitting the registered-population event study by treated size group shows the response in both treated groups, most precisely for cities of 0.5–1 million (coefficients of 0.067 to 0.109 in 2014–2016 with flat pre-trends), with later and noisier estimates for cities below 0.5 million.

3.2 Per Capita Unrest Rates

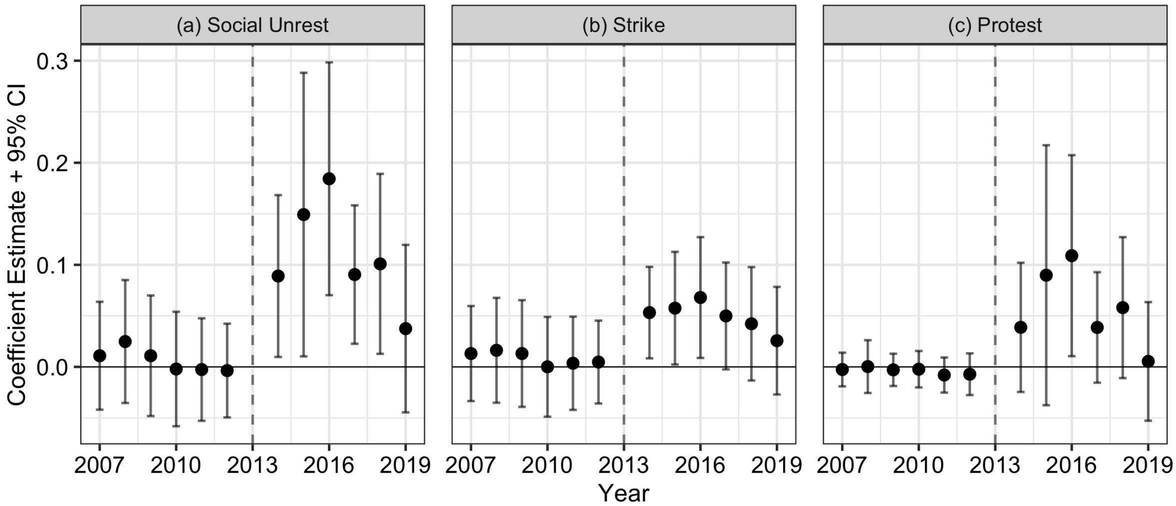
We report the per capita unrest rates (events per 100,000 registered population) in columns (1)–(3) of Table 1. The time-varying registered population is the primary and policy-relevant denominator, given that the Hukou reform led to an increase in registered populations in treatment cities (Section 3.1). Because the denominator itself grows with the reform, dividing by an expanding population causes the estimated per-capita increase to understate what a fixed-population baseline would show. In columns (4)–(6) of Table 1 we make this point more concrete. Replicating the specification with 2010 registered population as a fixed denominator (column 4) yields estimates of 0.139 for overall unrest, 0.054 for strikes (column 5), and 0.082 for protests (column 6), all larger than the corresponding time-varying estimates of 0.104 (column 1), 0.042 (column 2) and 0.060 (column 3). The gap is the share of the per-capita increase that the growing denominator absorbs. Far from undermining the main result, the fixed-denominator columns confirm that unrest rose faster than the expanding registered population, and that the time-varying estimates are conservative lower bounds. The fixed-denominator columns also serve as a direct comparison with Lai and Qiu (2025), who normalize by pre-reform prime-age physical population.[10]

In Figure 3 we present the event study estimates of Eq. (1) for per capita unrest rates. The pre-reform coefficients range between -0.004 and $+0.025$ events per 100,000, with p-values from 0.42 to 0.94, and the same pattern holds for strikes and protests, supporting the parallel trends assumption.[11] After 2014, per capita unrest increased in treatment cities relative to control cities. The overall effect builds from 0.089 events per 100,000 in 2014 to a peak of 0.184 in 2016 ($p < 0.01$), about 4.5 times the treated group's pre-reform mean of 0.041 events per 100,000. Column (1) of Table 1 confirms a sustained differential of 0.104 events per 100,000 ($p < 0.01$), about 2.5 times the treated pre-reform mean. These magnitudes are the reform-attributable differentials relative to control cities. Both groups experienced a larger common increase after 2013, which the design nets out. The fixed-denominator estimate in column (4) is 0.139 ($p < 0.01$), confirming the attenuation interpretation. Effects remain elevated through 2018 before declining toward zero in 2019. Both strikes and protests contribute, with rates elevated throughout 2014–2018 in panels (b) and (c) of Figure 3. The attenuation after 2016 is consistent with labor market adjustment as migrants are sorted into accessible employment over time, a channel we explore

[10] Count-based (PPML) estimates show a positive but smaller differential than the per-capita specification, for the mechanical reason that a single additional event in a treatment city of 600,000 contributes 0.17 events per 100,000 residents while the same event in a control city of two million contributes only 0.05.

[11] The parallel trends pattern is robust to adding city-specific linear time trends (Section 3.4, panel c).

Figure 3: Per Capita Unrest Rates (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for per capita unrest rates (events per 100,000 registered population). Panel (a) reports overall labor unrest, panel (b) reports strikes, and panel (c) reports protests. Sample, specification, and reference year as in Figure 2.

further in Section 4.

Figure 4 extends the event study to all five city-size groups, using megacities as the common reference. We find that the post-reform pattern is monotone in reform intensity. Size 5 cities (below 0.5 million, complete elimination of requirements) show the largest increase in per capita unrest, size 4 cities (0.5–1 million, most extensive reduction) show a moderate and significant increase, size 3 cities show no detectable change, and size 2 cities are indistinguishable from megacities throughout the sample period. The gradient is sharpest for strikes. The monotone ordering across five distinct size groups is consistent with treatment intensity driving the differential response, though any confounder that loads monotonically on city size could in principle generate a similar gradient.[12]

In contrast, Lai and Qiu (2025) use a difference-in-discontinuity design around the 3 million population threshold and report a 42 percent reduction in labor unrest. The two designs test different margins of the same reform. Their 3 million cutoff compares cities that received minimal barrier reduction (1–3 million) against cities that received essentially none (3–5 million). Our 1 million cutoff compares cities that received the most extensive barrier reduction (below 1 million) against cities with only partial reduction. Using our data, the same 3 million cutoff does not reproduce their result. When we ap-

[12]We find that some early-year coefficients are absent for size 5 cities because the size-5 year interaction is collinear with the province-by-year fixed effects in sparse province-year cells. This affects only the pre-2011 years, when CLB coverage of the smallest cities was extremely thin (3–5 total events per year across all 51 size-5 cities), and does not affect the post-reform estimates.

Table 1: Difference-in-Differences Estimates for Unrest Outcomes

	Time-Varying Population			Fixed 2010 Population		
	Unrest	Strike	Protest	Unrest	Strike	Protest
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × Post	0.104***	0.042***	0.060**	0.139***	0.054***	0.082**
	(0.033)	(0.010)	(0.028)	(0.038)	(0.010)	(0.033)
Dep. var. mean	0.049	0.040	0.010	0.051	0.041	0.011
Observations	3,487	3,487	3,487	3,461	3,461	3,461
R^2	0.64	0.48	0.61	0.66	0.49	0.64

Notes: All columns report per capita unrest rates (events per 100,000 registered population). Columns (1)–(3) use time-varying registered population as the denominator. Columns (4)–(6) use 2010 registered population as a fixed denominator. All columns are estimated by OLS. Post = 1 for years after 2013. Dep. var. mean reports the pre-reform sample mean (2007–2013) for columns (1)–(3). Sample and specification as in Figure 2. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

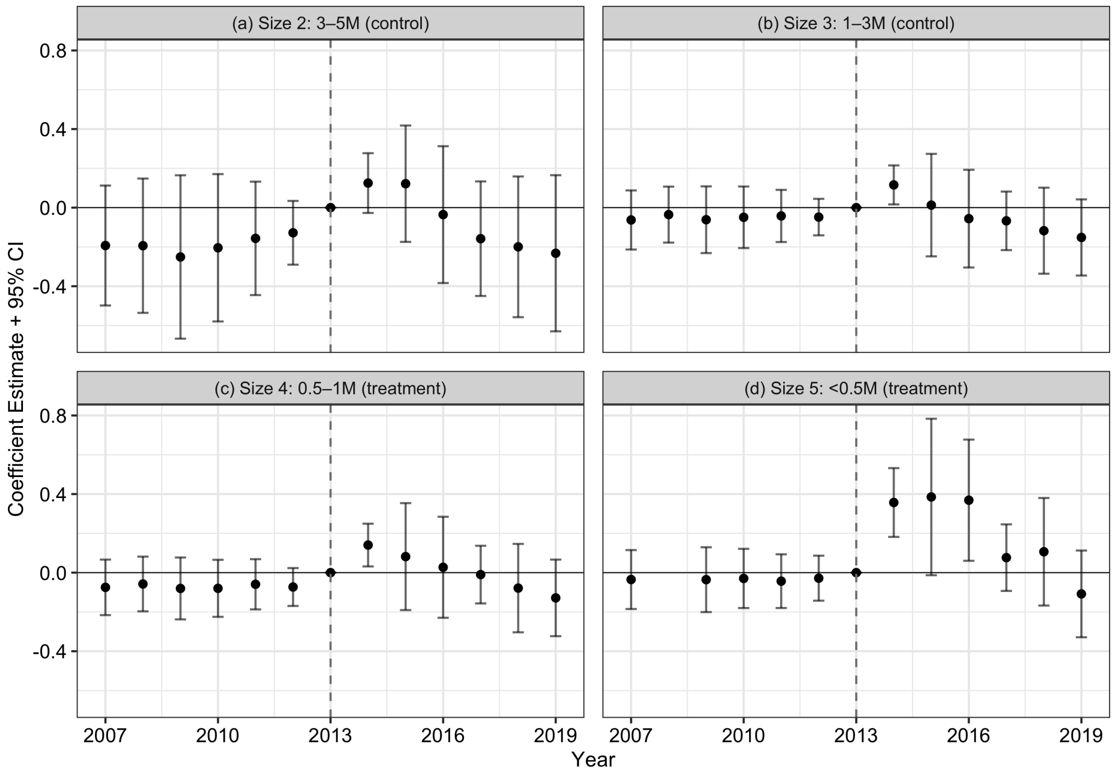
ply this cutoff to our sample, the estimated coefficient is -0.045 ($p > 0.3$), with only 11 control cities and standard errors roughly 60 percent larger than our baseline (Table 2). Appendix Figure A2 shows that the event-study coefficients are imprecise and exhibit no clear break at 2013, consistent with no detectable effect at this cutoff. The null here and the sharp by-size gradient in Panel B of Table 2 are two sides of the same pattern rather than a contradiction. The 3 million split places the heavily and the barely treated cities on the same side of the cutoff and leaves 11 cities on the other, so it averages the intensity gradient away and loses the power to detect it.

The divergence between the two papers reflects differences in identification strategy, treatment definition, and outcome measurement. Lai and Qiu (2025)'s difference-in-discontinuity design relies on a polynomial in the running variable to absorb the relationship between city size and unrest dynamics, an assumption sensitive to functional form. Their single treatment indicator pools 250 reform cities spanning our entire treatment and control groups, masking dose-response heterogeneity. Furthermore, their outcome denominator is prime-age physical population, which, as we discussed earlier, did not change after the reform. In contrast, our denominator variable is the registered population, which captures the formalization margin the reform directly affected.[13]

We can also narrow the control group in the opposite direction, using only cities with

[13] Without the polynomial control, reform and non-reform cities in Lai and Qiu (2025) show significant pre-existing differences in unrest growth, population growth, and migrant shares (their Table 2), so identification depends on the polynomial correctly absorbing these differences. Our panel provides six pre-reform years of parallel trends evidence compared to two in their design.

Figure 4: Per Capita Unrest by City Size Category (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for overall labor unrest per 100,000 registered population, using megacities (size 1, population above 5 million) as the reference group. Panel (a) shows size 2 cities (3–5 million), panel (b) shows size 3 cities (1–3 million), panel (c) shows size 4 cities (0.5–1 million), and panel (d) shows size 5 cities (below 0.5 million). Some early-year coefficients for size 5 are absent when absorbed by province-by-year fixed effects in sparse cells. Fixed effects, clustering, and reference year as in Figure 2.

population 1–3 million as controls for our treatment cities. This drops the largest control cities (3–5 million) and asks whether our main results hold against a more homogeneous comparison group. Table 2 shows that they do. The estimated effect increases monotonically with reform intensity, 0.071 per 100,000 for cities with population 0.5–1 million ($p < 0.05$) and 0.231 for cities below 0.5 million ($p < 0.01$). Appendix Figure A3 shows broadly flat pre-trends and a post-reform increase for both treatment groups, with larger effects for the smallest cities. The monotone gradient is what the reform predicts. The sectoral composition of the per capita increase, to which we now turn, provides evidence on the labor market channel driving this result.

Table 2: Alternative Population Cutoffs

	Unrest (1)	Strike (2)	Protest (3)
Panel A. Treatment = Below 3 Million, Control = 3–5 Million			
<3M × Post	-0.045	0.045	-0.087
	(0.052)	(0.032)	(0.066)
Observations	3,487	3,487	3,487
R^2	0.63	0.47	0.61
Panel B. Treatment = Below 1 Million by Size, Control = 1–3 Million			
0.5–1M × Post	0.071**	0.030***	0.042
	(0.033)	(0.009)	(0.028)
<0.5M × Post	0.231***	0.070***	0.154***
	(0.060)	(0.019)	(0.049)
Observations	3,344	3,344	3,344
R^2	0.63	0.44	0.61

Notes: All columns report per capita unrest (events per 100,000 registered population). Each panel re-estimates the pooled DiD with an alternative split of the sample by 2013 urban population. Panel A treats all 262 cities below 3 million as one group and compares them with the 11 cities of 3–5 million. Panel B reports separate effects for cities of 0.5–1 million and cities below 0.5 million, compared with the 108 cities of 1–3 million. The contrast between the panels reflects the design. Panel A pools heavily and barely treated cities and relies on 11 comparison cities, while Panel B aligns the comparison with treatment intensity. Post = 1 for years after 2013. Fixed effects and clustering as in Figure 2. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.3 Sectoral Heterogeneity

A distinctive feature of the CLB incident database is that each record identifies both the industrial sector and the participant count in each incident. This enables a decomposition of the aggregate per capita effect by the type of unrest. Both the sector and participant-count fields are recorded directly in the CLB data, and the grouping into four sectors is the authors' aggregation. We classify the raw CLB sector categories into four groups, Manufacturing (manufacturing, heavy industry, and mining), Construction (construction), Transportation (transportation, storage, logistics, and postal services), and Services (retail, education, and the public sector). This decomposition is our migrant-exposure test. We hypothesize that if the Hukou reform operated through migrant workers rather than a city-size shock, unrest should rise in sectors with stronger migrant worker presence than others. We adopt two sector classifications in the paper and they serve different purposes. The CLB sectoral classification described above is narrow, with transportation as its own category, because it follows the sectors recorded in each incident. The employment analysis in Section 4 uses the Statistical Yearbook's secondary and tertiary aggregate classification, in which the tertiary sector subsumes transportation along with wholesale and retail trade, social services, finance, and health. Furthermore, the CLB records participant counts in pre-coded bins (1–100, 101–1,000, 1,001–10,000, and 10,000+). We follow the CLB coding breaks and classify events with 1–100 participants as small-scale and events with 100 or more participants as large-scale. In Figure 5 we present the per capita event study estimates for the eight categories that are created by the intersection of sectors and incident scales, using the same fixed effects and sample coverage as the baseline specification. Appendix Table A7 reports pooled DiD estimates for all eight sector-incident scale categories.

The per capita increase in unrest is concentrated in manufacturing, construction, and transportation, the three sectors migrants most often enter.[14] Worker unrest events in small-scale manufacturing exhibit a sharp post-reform increase in the event-study estimates, with coefficients rising from near zero in 2013 to around 0.04–0.09 events per 100,000 population in 2015–2016. In addition, worker unrest events in large-scale manufacturing display a more persistent positive differential, between 0.03 and 0.05, throughout the post-reform period.[15] Construction unrest incidents also rose, with small-scale construction events showing a sharp post-reform increase to approximately 0.09 per 100,000 in 2015–2016.[16] Transportation incidents show a more modest but consistent positive post-reform differential. Appendix Table A7 reports the pooled DiD estimates for all eight sector-scale categories. The estimates are positive and statistically significant in the three migrant concentrated sectors on both incident scales, with the single exception of large-scale construction, where major disputes (100+ participants) are rare in smaller cities. Both service categories are indistinguishable from zero.

The sector-specific event rates in Figure 5 use total registered population as the denominator, so they measure resident exposure to unrest in each sector. Sector composition matters if treatment cities shift toward the relevant sector after the reform: sector incidents per resident could rise even if incidents per sector worker did not. Appendix Figure A4

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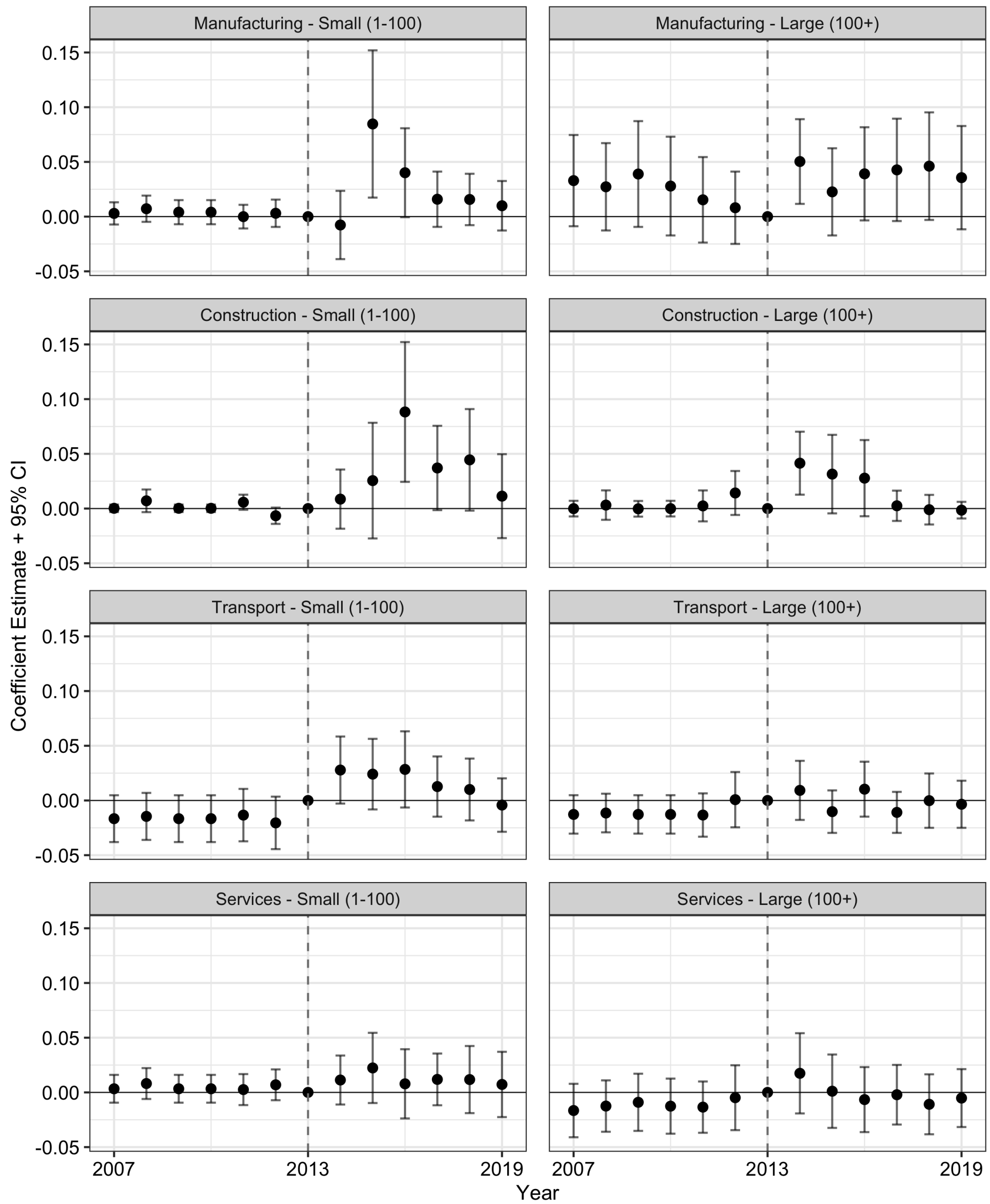
[14]The National Bureau of Statistics Migrant Worker Survey, covering 31 provinces and approximately 1.5 million observations, reports that manufacturing and construction together provide 52–55 percent of employment for migrant workers nationally, making them the two largest entry sectors (National Bureau of Statistics of China, 2015). Transport, storage, and postal services account for an additional share.

[15]The Manufacturing–Large panel in Figure 5 is positive in every pre-reform year. The coefficients range from 0.008 to 0.039 events per 100,000 residents, and none is individually distinguishable from zero. The Manufacturing–Small panel has a cleaner flat pre-trend and provides the sharper event-study evidence.

[16]Appendix Table A8 shows that construction events are sparse before electronic CLB coverage begins in 2011. Small-scale construction appears only once in 2007–2010, in treatment cities in 2008, so the earliest construction coefficients are estimated from very limited raw variation.

Figure 5: Per Capita Unrest by Sector and Incident Scale (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for per capita unrest (events per 100,000 registered population) by sector and incident scale. Rows show sectors and columns show incident scale. Manufacturing covers manufacturing, heavy industry, and mining. Construction covers construction. Transport covers transport, storage, logistics, and postal services. Services covers retail, education, and the public sector. Small = 1–100 participants. Large = 100+. Sample, specification, and reference year as in Figure 2.

and Table A9 check this possibility by rescaling manufacturing and construction incidents by secondary-sector employment, and transport and service incidents by tertiary-sector employment. The sectoral pattern remains unchanged. Manufacturing incidents remain positive on both scales (0.398 small-scale and 0.187 large-scale per 100,000 secondary workers, both $p < 0.10$), as do small-scale construction incidents (0.724, $p < 0.10$) and small-scale transportation incidents (0.205, $p < 0.01$). Both service categories and large-scale construction incidents remain indistinguishable from zero.

These findings raise the question of which individuals participated in these events. The CLB does not record the Hukou status of individual participants and therefore we do not observe the participant-level registration status. However, we can characterize the affected workers indirectly through three pieces of evidence. First, the grievance field shows that wage arrears are the primary listed cause. Second, the sector field shows that incidents concentrate in manufacturing, construction, and transportation, the three sectors where migrant employment concentrates nationally. Third, the individual-level survey evidence in Section 4.3 shows that wage decreases fell on migrants in the same sectors, concentrated among longer-tenure incumbents rather than recent arrivals. Together the grievance, sectoral, and survey evidence point to incumbent migrant workers in migrant-intensive sectors as the main drivers of unrest events.

3.4 Robustness Analysis

The baseline per capita finding could reflect sample selection bias, concurrent shocks and omitted variable bias, measurement error, or data limitations rather than the Hukou reform itself. We address four categories of concern in turn in this section, alternative sample restrictions and confound controls, estimator and denominator choice, an alternative control group (megacities), and differential CLB reporting.

Alternative specifications. We report per capita unrest estimates under eight alternative specifications in Table 3. Panels (a) and (b) restrict the sample. Panel (a) excludes autonomous minority regions (Tibet, Xinjiang, Guangxi, Inner Mongolia, and Ningxia), where unique political and societal factors could drive labor unrest independent of Hukou policy changes. Panel (b) restricts the sample to 2011–2019, the period with consistent electronic CLB coverage, dropping the pre-2011 years reconstructed from annual reports. The estimates remain virtually identical to the baseline in both cases. Panel (c) adds city-specific linear time trends, addressing gradual differential city trajectories that may not be fully captured by pre-trend tests. These trends also help account for slowly evolv-

Table 3: Pooled DiD Estimates Under Alternative Specifications

Specification	Treat × Post (SE)	N
(a) Drop autonomous minority regions	0.110*** (0.035)	3,096
(b) Restrict to 2011–2019 (electronic CLB coverage)	0.111*** (0.032)	2,456
(c) Add prefecture linear trends	0.107*** (0.034)	3,487
(d) Add trade openness	0.108*** (0.032)	3,475
(e) Add environmental policy	0.107*** (0.032)	3,475
(f) Add PM2.5 concentration	0.111*** (0.033)	3,414
(g) Control supply-side reform (2010 sec. share × post-2015)	0.103*** (0.033)	3,461
(h) Control SOE share (2010 urban unit share × post)	0.105*** (0.033)	3,461
(i) Year FE only (no province × year)	0.109*** (0.031)	3,487

Notes: Each row reports the pooled DiD estimate (Treat × Post) for per capita overall unrest (events per 100,000 registered population). N refers to city-year observations. Standard errors in parentheses. Unless a row states otherwise, sample and specification as in Figure 2. Specification (g) interacts each city's 2010 secondary employment share with an indicator for years from 2015 onward. Specification (h) interacts the 2010 urban unit employment share with the post-reform indicator. Specification (i) replaces province-by-year fixed effects with year fixed effects only. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

ing city-level labor-market conditions, including differential minimum wage trajectories that can vary across cities within a province. The estimated effect remains positive and statistically significant, though slightly attenuated.

Panels (d) through (h) address specific concurrent shocks that could confound the baseline estimate. Panel (d) controls for city-level trade openness, since China's export slowdown after 2014 differentially affected manufacturing-heavy smaller cities and could independently generate labor unrest. Panels (e) and (f) address environmental regulation. Panel (e) adds interaction terms between year fixed effects and an indicator for the "2+26" cities subject to intensified environmental monitoring and factory shutdowns post-2014,[17] and panel (f) adds city-level PM2.5 concentrations (Khanna et al., 2025; Li and Meng, 2023), since pollution-related protests are a documented form of collective action in China. Panel (g) controls for the supply-side structural reform campaign that began in 2015, which shut excess capacity in coal, steel, and cement and could have generated layoff-driven unrest in manufacturing-heavy cities, by interacting each city's 2010 secondary employment share with an indicator for years from 2015 onward. Panel (h) addresses the anti-corruption campaign that intensified after 2012, which disrupted patronage networks and could provoke protest in cities with larger state-owned enterprise presence, by interacting the 2010 urban unit employment share with the post-reform in-

[17] The "2+26 Cities" refers to a regional air-quality management strategy that includes Beijing, Tianjin, plus 26 nearby prefecture-level cities in northern China (Shu et al., 2022).

dicator. Panel (i) replaces province-by-year fixed effects with year fixed effects only. The estimate remains virtually unchanged (0.109 versus 0.104 in the baseline), confirming that the result does not depend on the more demanding fixed effect structure. Across all nine specifications the estimated DiD coefficient remains positive, statistically significant, and close to the baseline.

Inference. Appendix Table A4 reports joint F-tests of pre-reform event study coefficients. For all headline specifications, the null that pre-reform coefficients are jointly zero cannot be rejected ($p = 0.42$ to 0.88). For the manufacturing-large subcategory, the F-test yields $p = 0.11$, consistent with the slightly noisier pre-reform patterns visible in the event study for that panel. The same test applied to the registered-population and sectoral-employment event studies that anchor the mechanism analysis also fails to reject (registered population $p = 0.16$, secondary employment $p = 0.33$, tertiary employment $p = 0.74$). The visual pattern in which every pre-period point estimate sits above the 2013 reference value in those figures therefore reflects variation around the normalized year, not statistically significant pre-trends. Appendix Figure A1 presents a sensitivity analysis following Rambachan and Roth (2023). The confidence interval for the pooled post-treatment effect excludes zero when post-treatment violations of parallel trends are permitted to be up to half as large as the maximum pre-reform violation ($\bar{M} = 0.5$), and includes zero at $\bar{M} = 1.0$.

Alternative estimator. Per capita unrest rates are non-negative and right-skewed. OLS may therefore impose an inappropriate linear conditional mean. Appendix Table A10 shows that pooled DiD estimates remain positive and statistically significant for overall unrest and strikes under PPML estimation, while the protest estimate becomes imprecise. Overall, the directional agreement in point estimates confirms that the OLS results are not an artifact of the linear specification.

Alternative control groups. We compare treatment cities with megacities (population above 5 million) rather than with control cities, an entirely different comparison group that received no barrier reduction. Appendix Figure A5 shows the event-study dynamics and Appendix Table A11 reports the pooled DiD estimates. The results are consistent with the baseline. Strikes are positive and statistically significant under both time-varying and fixed denominators (0.066 and 0.087, both $p < 0.01$). Overall unrest is positive but less precise (0.103 and 0.177, the latter significant at the 10 percent level), because, as discussed in Section 2.3, megacity protest activity rose sharply in 2015–2016 alongside the stock market crash and the intensified anti-corruption campaign, and those shocks enter the counterfactual when megacities serve as controls. The registered-population response against this benchmark is similar in magnitude to the baseline first stage but imprecisely estimated.[18] We read this comparison as the same mechanism measured against a weaker benchmark. With only 12 megacities that differ from smaller cities in political status and economic structure, the comparison has less power and less comparability than the baseline contrast between the two reform arms, and the consistency of the strike results indicates that the baseline finding is not an artifact of the control comparison.

Data validity. As discussed in Section 2, the CLB's coverage expanded substantially over the sample period, with smaller cities gaining disproportionately from the shift to electronic collection after 2011. If reporting improvements were concentrated in treatment cities, the post-reform increase could partly reflect better detection rather than more unrest. Three features of the results make pure reporting bias a less complete explanation. First, the CLB is a nongovernmental organization based in Hong Kong that compiles incidents from social media, overseas Chinese media, and labor activists (Qin et al., 2024; Elfstrom and Kuruvilla, 2014). Its coverage, therefore, does not depend on local government cooperation or reporting incentives. Second, the sectoral decomposition shows positive differentials for both small-scale (fewer than 100 participants) and large-scale (100 or more participants) manufacturing and construction incidents. Large collective actions attract regional and national media attention regardless of city size, so differential social media penetration in smaller cities cannot explain the large-scale pattern. A detection-bias mechanism would need to generate spuriously elevated counts for both scale categories simultaneously, while leaving service sector counts unaffected. Third, the individual-level CMDS wage evidence, which comes from an independent government survey unaffected by CLB reporting changes, shows migrant wage declines concentrated in the same sectors where unrest rose. Nevertheless, reporting bias remains a concern that we cannot fully rule out with our research design.

4 Mechanisms

The concentration of post-reform unrest in manufacturing, construction, and transportation raises a mechanism question. Why these sectors and not others? The population evidence narrows the possible explanations. The registered population rose 6–11 percent

[18] The registered-population differential against megacities is 0.057 log points (s.e. 0.077), close to the baseline estimate of 0.058.

while the physical population showed no detectable change (Section 3.1), thus the reform's dominant margin was granting formal status to workers who were already present, with at most a modest inflow of new arrivals. Accounts that require a large physical inflow, including native backlash against newcomers and physical congestion of roads, schools, and hospitals, are hard to square with this evidence. The margin the reform moved was primarily institutional. Workers living and working in treatment cities acquired formal local residency, and the same cities became easier for new migrants to enter.

Our account combines three reinforcing channels; i) a gap between entitlements and delivery, ii) congestion in the sectors migrants can enter, and iii) a lower cost of collective action under permanent residency. Hukou registration entitles its holder to local social insurance and public services and confers a permanent stake in the destination city. It does not change labor-contract or dispute-resolution rights, which the 2008 Labor Contract Law extends to all employees regardless of registration status. Formalization therefore raised what workers could legitimately expect from employers and local governments without changing the jobs they held. Two observable gaps anchor the account. New migrants' social insurance coverage fell by 2.1 percentage points in treatment cities after the reform (Appendix Table A12), thus employment did not deliver the insurance enrollment that Hukou registration promised. And wage arrears, the failure of employers to pay what was already owed, are the primary grievance in 74 percent of CLB events with identified causes. The wage-setting evidence in Guo et al. (2025), who show that Hukou eligibility enters the worker-employer wage bargain directly as a compensating differential within firms, provides evidence for why a change in registration status alone can move workplace conflict.

This account rests on three assumptions, each with empirical support. The first is behavioral. Workers judge what they receive against what their formal status entitles them to, and they act when the pay gap exceeds the cost of acting. Formal status moves both terms, raising the entitlement and lowering the cost of action through the permanent residency rights. The second is technological. Because capacity in the accessible sector(s) is fixed in the short run, the Hukou reform did not increase either the stock of manufacturing positions or the volume of freight and passengers over which transport workers compete. Consequently, realized earnings could not rise to match the expansion in entitlements. The third is institutional. Credentialed service occupations remained closed to migrant entry and paid above the entitlement; thus no shortfall could form there. Under these assumptions the direction of the reform's effect is conditional rather than universal, which is what reconciles our finding with the legalization literature. Formalization lowers unrest where delivery rises to meet the new entitlements, and raises unrest where entitlements outrun delivery. And because the entitlements increase and the cost of acting falls for workers already present, the effect does not depend on a physical inflow, consistent with the evidence that formalization of incumbents was the dominant margin driving our results.

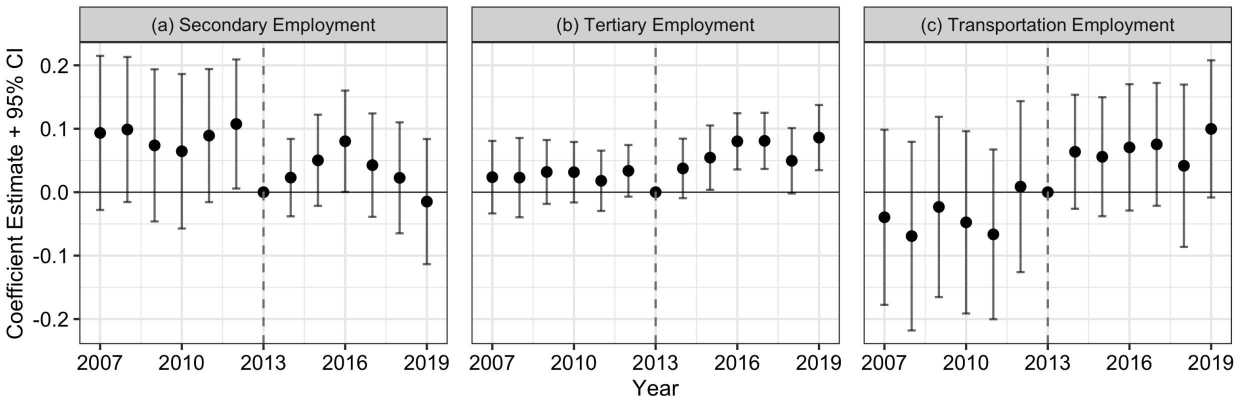
A second subordinate account is congestion within the sectors that migrants can enter. A queuing logic in the tradition of Harris and Todaro (1970) and Fields (1975) implies that when more workers seek a fixed stock of positions, congestion rises even if wages are rigid. The competition in this account comes from newly formalized workers reallocating across sectors, joined by whatever modest inflow the reform induced; thus it does not require a large physical inflow. We argue that the two accounts reinforce each other. Both predict increasing grievances where the gap between formal status and actual conditions is largest and both predict concentration in manufacturing, construction, and transportation, where informal and piece-rate arrangements are most prevalent.

These accounts generate four testable predictions. First, secondary-sector employment should show no differential expansion in treatment cities. Second, tertiary-sector growth should concentrate in accessible service margins rather than expand broadly. Third, the gap between formal entitlements and realized conditions should widen in the sectors where unrest rose, and it should surface through each sector's pay structure. Manufacturing's rigid wage floor pushes the shortfall into job rationing and unpaid wages, while transport's flexible piece rate pushes it into earnings, so the same gap should appear as arrears in manufacturing and as earnings compression in transportation services. Fourth, the effect should be strongest during the initial adjustment period and attenuate as workers and employers adapt. If instead unrest rose uniformly across all sectors, or if manufacturing expanded alongside the formalization wave, or if service employment grew broadly across all subsectors, these explanations would be much less plausible. Employment and wages are joint equilibrium outcomes, so we read the sectoral patterns as tests of these predictions rather than as direct measures of labor demand. We examine employment growth, wages, and fiscal capacity against each prediction in the next subsection.

4.1 Sectoral Employment Responses

Figure 6 presents event study estimates for (log) employment in the secondary sector (which includes manufacturing), the tertiary sector (services), and the transportation subsector within services, comparing treatment and control cities. The two sectors tell sharply

Figure 6: Employment Responses by Sector (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for log employment. Panel (a) reports secondary-sector employment, panel (b) reports tertiary-sector employment, and panel (c) reports transportation and communication employment, the tertiary subsector with the clearest post-reform break. Joint F-tests cannot reject zero pre-reform coefficients in panels (a) and (b) (Appendix Table A4). Sample, specification, and reference year as in Figure 2.

different stories. In panel (a), secondary employment shows no clear post-reform break. The pre-reform coefficients are positive and somewhat uneven, though the joint pre-trend test does not reject zero ($p = 0.33$, Appendix Table A4). The post-reform coefficients remain near zero with wide confidence intervals, and the pooled DiD estimate is -0.041 (Appendix Table A13), statistically indistinguishable from zero. The reform did not generate additional differential growth in secondary employment. Manufacturing could not absorb incoming workers at a faster rate than it had in the years leading up to the reform.

Tertiary employment responded differently. Panel (b) shows a clear positive break after 2013, with coefficients rising to 0.05–0.09 by 2016–2019. The pooled DiD estimate of 0.042 ($p < 0.10$, Table A13) corresponds to approximately 4 percent higher service employment in treatment cities after the reform. Service employment expanded; manufacturing did not. Panel (c) previews where that growth concentrates. Transportation employment breaks cleanly upward in 2013 and we examine this pattern in Section 4.2 in detail. The reform attracted population to smaller cities but created additional jobs primarily in services, leaving secondary-sector capacity essentially unchanged. This asymmetry clarifies the denominator robustness check in Appendix Table A9: manufacturing unrest remains positive when scaled by secondary employment, while service unrest remains near zero when scaled by tertiary employment.

4.2 Labor Absorption in the Service Sector

The aggregate tertiary employment expansion documented above leaves open the question of which service channels absorbed the incoming workers. The distinction matters for the congestion account. If growth concentrated in sectors with low entry barriers, such as transportation and logistics, where a newly arrived worker can quickly find employment without credentials or substantial employer investment, then the sector absorbs migrants at the pace they arrive. If instead growth concentrated in subsectors requiring physical capital investment or formal qualifications, incoming migrants would face congestion in the limited positions they can actually fill. Appendix Figure A6 decomposes tertiary employment into six subsectors.

Among the six subsectors, transportation and communication is the only one showing a statistically significant differential expansion (column 4, Appendix Table A13). The pooled DiD estimate of 0.101 ($p < 0.10$) corresponds to approximately 10 percent higher transportation employment in treatment cities after the reform, or roughly 540 additional workers per city against a pre-reform base of about 5,400 ($\exp(8.60)$). The event study for transportation shows a clean break in 2013, with coefficients rising from near zero to 0.07–0.10 in the post-reform years, and no indication of pre-reform divergence. Wholesale and retail trade shows a directionally positive post-reform trend in the event study (coefficients reaching 0.07–0.13 before moderating), but the pooled estimate of 0.071 is not statistically significant. Healthcare (0.081) and scientific and technical services (0.038) display gradual upward movement in the post-reform event study, though neither approaches conventional significance levels. Finance (0.018) is essentially flat throughout, and social services (0.040) exhibits large pre-reform swings followed by convergence toward zero, a pattern more consistent with structural catch-up than a reform response.[19]

The transportation sector presents an apparent paradox. Employment expanded by approximately 10 percent here, yet per capita unrest in this sector rose. One possible explanation is that transportation services absorbed additional workers without absorbing their earnings expectations. Transportation in Chinese cities operates predominantly through piece-rate arrangements, where drivers earn per trip or per delivery. More drivers competing for a given volume of freight or passengers mechanically compresses per-trip earnings even as total sector employment grows. CLB incident descriptions support this interpretation. Among the 499 transportation unrest events recorded in treatment cities during 2014–2019, 31 percent cite earnings grievances (fare disputes, wage arrears, fee

[19]Scientific and technical services shows some positive pre-reform drift in the event study. The post-reform trajectory for that subsector should therefore be interpreted cautiously.

complaints) and 46 percent cite competitive pressure (ride-hailing apps, unlicensed vehicles), with 68 percent matching at least one category.[20] The sector created positions while compressing the earnings attached to them, and the grievance composition shows that the compression was contested rather than absorbed. The individual-level wage evidence in the next subsection confirms the compression explanation directly.

These patterns reflect differential adjustment speeds across service sectors. Transportation, retail, and logistics are low-capital services with minimal credentialing requirements, so they can expand relatively quickly when population inflows arrive. Workers drawn into these accessible sectors are plausibly the same workers who would otherwise compete for manufacturing positions, as rural migrants to smaller cities typically seek employment in industry or basic services. Higher-skill services such as finance and healthcare require physical capital investment or regulatory approvals, restricting how quickly they can expand. Even if these sectors eventually grow, the workers they attract are unlikely to have been competing for manufacturing jobs in the first place. The net result is that accessible employment grew in a narrow set of service subsectors, insufficient to absorb the 6–11 percent registered population growth documented in Section 3.1. Workers who could not find positions in transportation or retail competed for manufacturing jobs, where secondary employment showed no differential expansion, contributing to the elevated per capita unrest documented in the preceding sections.

4.3 Migrant Wages and Sectoral Reallocation

The employment dynamics described above raise the question whether wage pressure and worker reallocation can be observed directly in individual-level data. To address this question, we use the China Migrants Dynamic Survey (CMDS) and examine both aggregate wages and sector-specific patterns in Eq. (2).[21] We follow the specification in An et al. (2024), who use the same survey to study migrant wage effects of the Hukou reform.

$$\ln w_{ict} = \alpha + \beta\,\mathrm{Treat}_c \times \mathrm{Post}_t + X'_{ict}\gamma + \mu_c + \delta_{p(c),t} + \phi_{o(i),t} + \varepsilon_{ict} \tag{2}$$

where w_{ict} is the monthly nominal wage of migrant i in city c at time t. X_{ict} is a set

[20] We classify incident descriptions using pre-specified keywords. Earnings terms cover fare, fee, deposit, unpaid wage, salary, income, and commission disputes; competition terms cover ride-hailing platforms, unlicensed vehicles, illegal taxis, and oversupply. Categories overlap because some events mention both earnings grievances and competition.

[21] The CMDS covers approximately 485,000 migrant workers across our sample cities from 2011 to 2017, providing only three pre-reform years (2011–2013) that limit the scope for pre-trends diagnostics on wage outcomes (Appendix Table A14 reports summary statistics).

of control variables and includes gender, age, age squared, education, marital status, and Hukou type. μ_c are city fixed effects, $\delta_{p(c),t}$ are province-by-year fixed effects, and $\phi_{o(i),t}$ are occupation-by-year fixed effects. Standard errors are clustered at the city level. We first estimate Eq. (2) pooling all sectors, then separately for manufacturing, construction, transportation, and other services to identify where wage pressure was concentrated. Panel B replaces the dependent variable with a sector indicator and uses the full sample of migrants with non-missing industry codes to test whether the reform shifted which sectors migrants work in.

Table 4 presents the results. Province-by-year fixed effects absorb province-level price trends in each year, so the identifying variation comes from within-province differences across city sizes net of common price movements. Panel A, column (1) pools all sectors and shows that migrant wages fell by 3.7 percent in treatment cities relative to control cities ($p < 0.01$). Columns (2)–(5) decompose the effect by sector. Wages fell significantly in construction (-4.1 percent, $p < 0.01$), transportation (-3.8 percent, $p < 0.05$), and other services (-4.1 percent, $p < 0.01$). These are the sectors where per capita unrest rose. Manufacturing wages declined by 1.7 percent but the estimate is not statistically significant (Appendix Table A15 confirms this result is stable across alternative specifications). These findings are consistent with An et al. (2024), who use the same survey and find migrant wage declines of 2.6–7.9 percent on average in non-megacities, with larger effects concentrated among less-educated workers in the smallest cities.

The non-significant manufacturing wage result is consistent with institutional wage floors that limit downward adjustment in this sector. Manufacturing wages in Chinese cities are subject to minimum wage floors and firm-level wage grids that compress downward adjustment. The grievance channel in manufacturing operates through employment access and working conditions rather than through wage levels. Registration promised newly formalized workers social insurance enrollment and a permanent stake in the city, yet employment conditions did not change. The CLB incident descriptions for manufacturing predominantly cite wage arrears and unpaid overtime rather than wage cuts, consistent with a formalization gap, where delivered conditions did not match what formal status implied.

Panel B asks whether the reform shifted which sectors migrants work in. The only significant reallocation is into other services (+1.7 percentage points, $p < 0.01$, on a pre-reform base of 44.1 percent). Manufacturing, construction, and transportation shares are unchanged (Appendix Table A15 confirms these results across alternative fixed effect structures). Migrants entered the most accessible service occupations, including delivery, catering, and resident services, where wages fell as a result. The cross-sector reallocation

Table 4: Migrant Wages and Sector Choice (2011–2017)

	All	Mfg.	Constr.	Transport	Other Svc.
	(1)	(2)	(3)	(4)	(5)
Panel A. Log wage					
Treat × Post	-0.037***	-0.017	-0.041***	-0.038**	-0.041***
	(0.009)	(0.012)	(0.015)	(0.015)	(0.012)
Pre-reform control mean	7.885	7.876	8.028	7.777	7.892
Observations	424,723	99,096	36,890	59,115	229,622
R^2	0.26	0.36	0.27	0.26	0.24
Panel B. Sector choice (= 1 if in sector)					
Treat × Post		0.003	-0.004	-0.001	0.017***
		(0.004)	(0.002)	(0.002)	(0.005)
Pre-reform control mean		0.246	0.082	0.121	0.441
Observations		485,496	485,496	485,496	485,496
R^2		0.61	0.62	0.64	0.53

Notes: Panel A estimates Eq. (2). Wages are in nominal values. Column (1) pools all sectors. Columns (2)–(5) restrict to migrants in the indicated sector. Panel B replaces the dependent variable with a sector indicator and uses the full sample. All specifications include individual controls (gender, age, age squared, education, marital status, Hukou type), city, province-by-year, and occupation-by-year fixed effects. Standard errors clustered at the city level. Data from the CMDS (2011–2017). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

margin parallels Guo et al. (2025), who document that changes in Hukou-quota access reallocate migrants across sectors and trace the wage consequences within employers. Their evidence captures the wage margin of that reallocation, while ours captures the unrest margin.

Appendix Table A16 decomposes the wage effect by migrant tenure. We find the wage decreases concentrated among longer-tenure migrants (incumbents already present when the reform took effect) rather than disproportionately affecting recent arrivals. The interaction between treatment and recent arrival status is insignificant in all four sectors. The reform did not protect incumbents from wage pressure, consistent with a broad congestion effect rather than a compositional shift driven exclusively by new entrants.

Appendix Table A17 reports city-level wages and fiscal capacity. Average city wages show no differential change (-0.006, s.e. 0.014, column 1), which is consistent with a composition difference between city averages and migrant-specific wages.[22] Per capita local

[22]Native wages showed no corresponding decline and were slightly positive (An et al., 2024), so in a city-level average the negative migrant effect and the flat native effect largely cancel.

public revenue declined by approximately 7 percent in treatment cities relative to control cities (-0.074, $p < 0.10$, column 2), as registered population grew faster than fiscal revenues. Per capita public expenditure also declined, though the estimate is smaller and imprecise (-0.052, column 3). The fiscal pattern matters for the formalization account because the reform expanded the population entitled to local services without a corresponding increase in per capita fiscal resources. Lower per capita revenue reduces local governments' capacity to enforce labor standards, staff labor arbitration courts, and fund the transitional public services that the reform promised to newly formalized residents. The negative but imprecise expenditure estimate suggests that spending did not offset this revenue pressure.

Beyond the grievances themselves, Hukou formalization may also have strengthened migrants' willingness to act on them. A worker with permanent local residency has a longer horizon in the city, a lower cost of remaining after a dispute, and less to lose from employer retaliation than a worker whose presence is administratively provisional. The insurance composition test (Appendix Table A12) measures the characteristics of new migrants rather than the position of incumbents, so the negative insurance result in column (4) (-2.1 percentage points, $p < 0.05$) leaves this stake effect among incumbent workers open.

The mechanism evidence we provide in this section establishes three stylized facts. First, secondary employment did not expand differentially in treatment cities. Second, the growth in tertiary employment concentrated in the transportation sector, a low entry-barrier sector. Third, migrant wages fell in sectors where unrest increased, while wage arrears dominated recorded grievances. These facts are consistent with a formalization gap, amplified by congestion in the narrow set of sectors that formalized workers could enter, and harder to reconcile with a story in which the reform expanded economic opportunity broadly. The evidence does not uniquely identify the relative importance of the formalization gap, sectoral congestion, and the stake effect. These channels are reinforcing rather than competing, and distinguishing among them would require worker-firm data on insurance enrollment, grievance filing, and payment practices that are not available at the city-year level.

5 Conclusion

We find that per capita labor unrest rose by about 2.5 times its pre-reform mean in the cities that received the most extensive barrier reduction under China's 2014 Hukou reform. The differential with control cities peaked in 2016 at about 4.5 times the pre-reform mean and concentrated in manufacturing, construction, and transportation sectors, where informal and piece-rate arrangements are most prevalent. The increase was monotonically graded across city-size categories, with cities receiving deeper barrier reductions showing larger per capita effects. The effect attenuated by 2019, a pattern consistent with labor market adjustment as workers and employers adapted over time, though the broader data-validity discussion cautions against over-interpreting trends in later years when CLB observability may also have shifted.

Taken together, the evidence is most consistent with a formalization gap. The reform granted formal status to migrant workers, most of them already present, expanding their entitlements to local social insurance and public services while the conditions of their employment, dominated by wage arrears, did not change. Congestion in the narrow set of accessible sectors reinforced the gap. Secondary employment did not expand, tertiary growth concentrated in a single low-barrier subsector, and migrant wages fell in the sectors where unrest rose. The reduced-form design cannot distinguish the relative importance of the formalization gap, sectoral congestion, and the stake effect of permanent residency, and these channels are reinforcing rather than competing. The principal design limitation is that treatment intensity is determined by city size; thus the estimates cannot fully separate the reform from other post-2014 shocks correlated with city size, even though the monotone dose-response pattern narrows the set of plausible confounders.

In terms of policy implications, the results imply that easing migration barriers does not mechanically reduce labor instability. The findings also support two narrower implications. First, formal status is not a substitute for delivery. When registration expands entitlements faster than employers and local governments deliver them, and the grievances in our data are overwhelmingly about unpaid wages, formalization itself can become a source of conflict. The contrast with immigrant legalization makes the condition precise. Formal status reduces conflict where market or legal pressure compels delivery to rise with it, and raises conflict where it does not. Second, the fiscal capacity of the receiving cities matters. Per capita public revenue fell as registered population grew, reducing the resources available to fund the services and labor-standards enforcement that the reform promised. The implication our findings can support is sequencing, pairing barrier reduction with the fiscal transfers and enforcement capacity needed to honor the new entitlements. Ranking the channels behind the unrest, and therefore choosing among more targeted responses, would require worker-firm data that the city-year design cannot provide.

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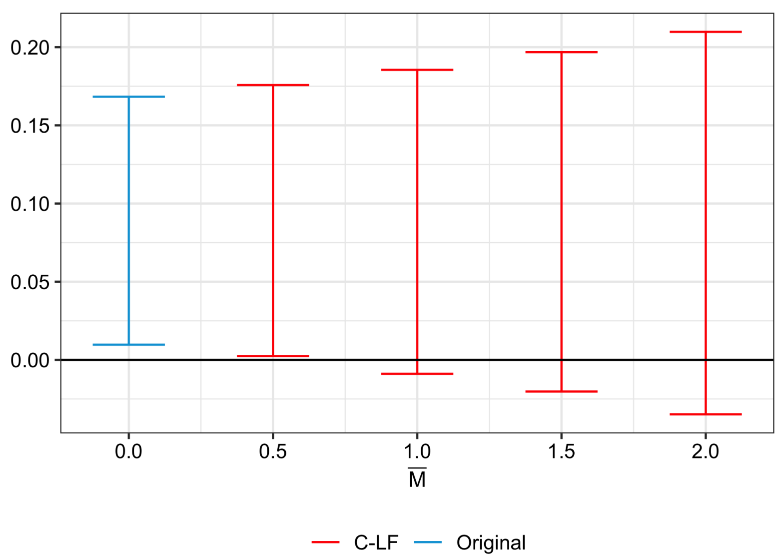
Online Appendix

Does Easing Migration Barriers Increase Labor Unrest? Evidence from the Hukou Reform in China

Unless a note states otherwise, every regression in this appendix compares treatment cities (sizes 4–5, population below 1 million) with control cities (sizes 2–3, population 1–5 million), excludes megacities (population above 5 million), includes city and province-by-year fixed effects, clusters standard errors at the city level, and uses 2013 as the omitted reference year in event studies.

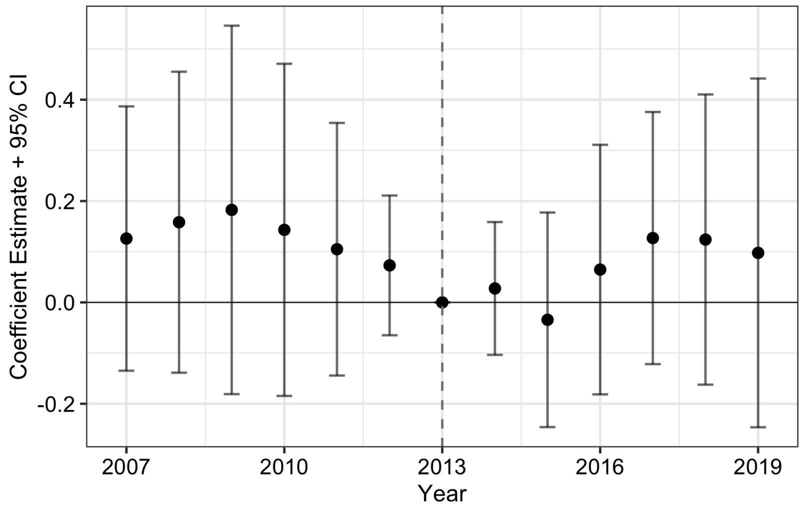
A1 Additional Figures

Figure A1: Sensitivity to Violations of Parallel Trends



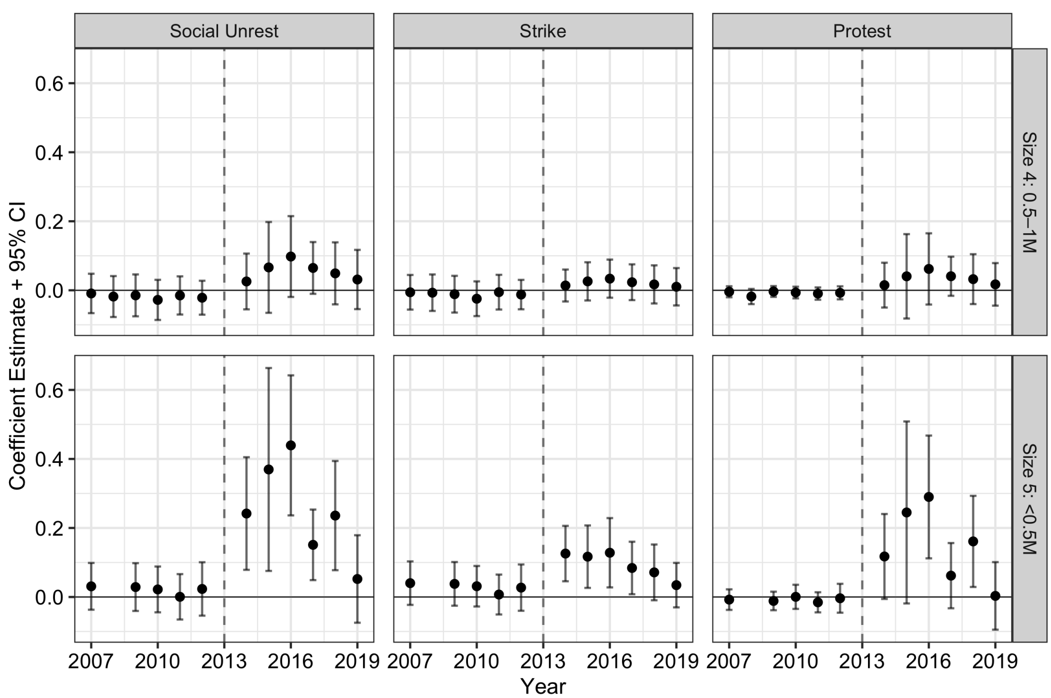
Notes: 95% confidence intervals for the pooled post-treatment effect on per capita overall unrest (events per 100,000 registered population) under the relative magnitudes approach of Rambachan and Roth (2023). Blue interval ($\bar{M} = 0$) assumes exact parallel trends. Red intervals allow post-treatment violations up to $\bar{M}$ times the maximum pre-treatment violation. Six pre-reform coefficients (2007–2012) and six post-reform coefficients (2014–2019).

Figure A2: Event Study, 3 Million Population Cutoff



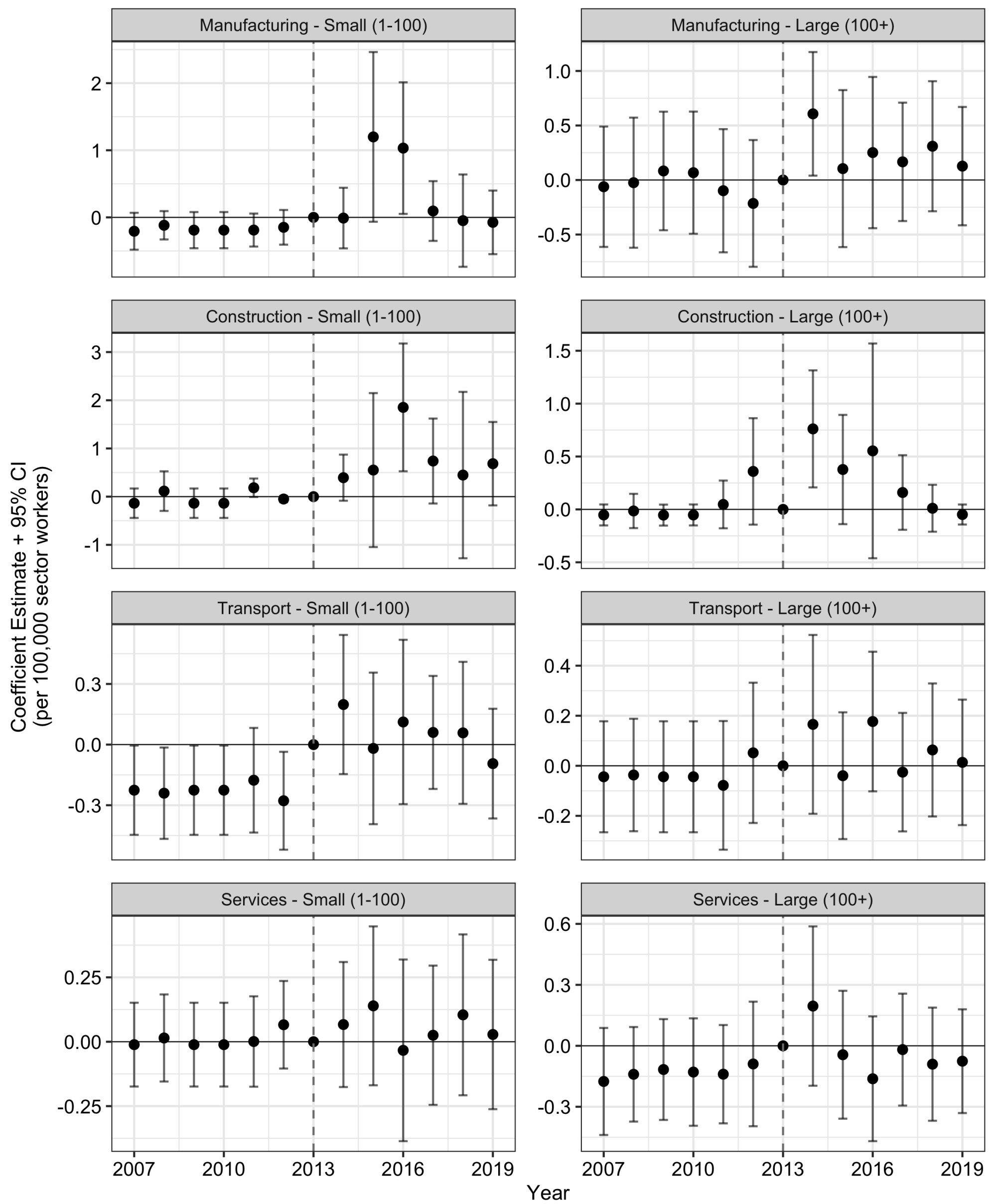
Notes: Event study coefficients and 95% confidence intervals for per capita unrest (events per 100,000 registered population) using a 3 million population cutoff. Treat = 1 for cities with population below 3 million (sizes 3–5, 262 cities). Treat = 0 for cities with population 3–5 million (size 2, 11 cities).

Figure A3: Event Study by City Size, Size 3 as Sole Control (1–3 Million Population)



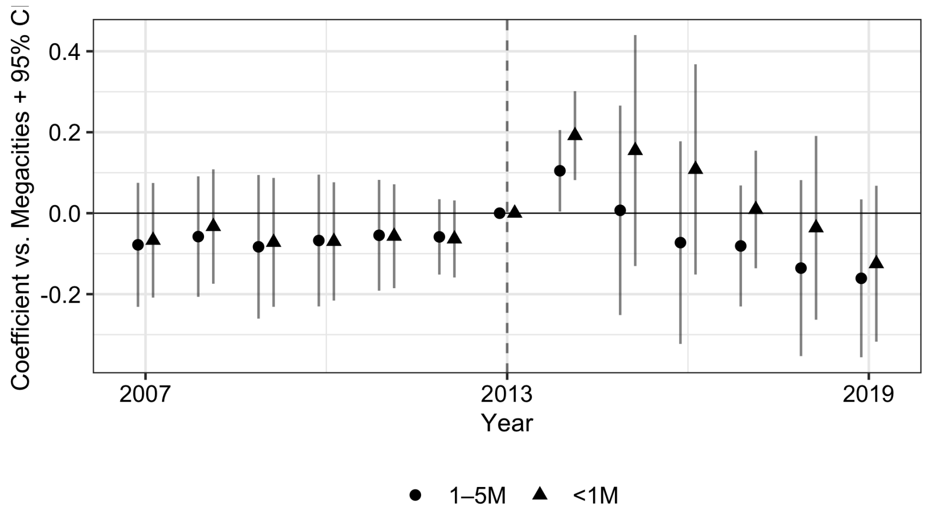
Notes: Event study coefficients and 95% confidence intervals using size 3 cities (population 1–3 million, 108 cities) as the sole control group. Rows show size 4 cities (population 0.5–1 million, 103 cities) and size 5 cities (population below 0.5 million, 51 cities). Columns show overall labor unrest, strikes, and protests, each per 100,000 registered population.

Figure A4: Unrest by Sector and Scale, Sector-Employment Denominator (2007–2019)



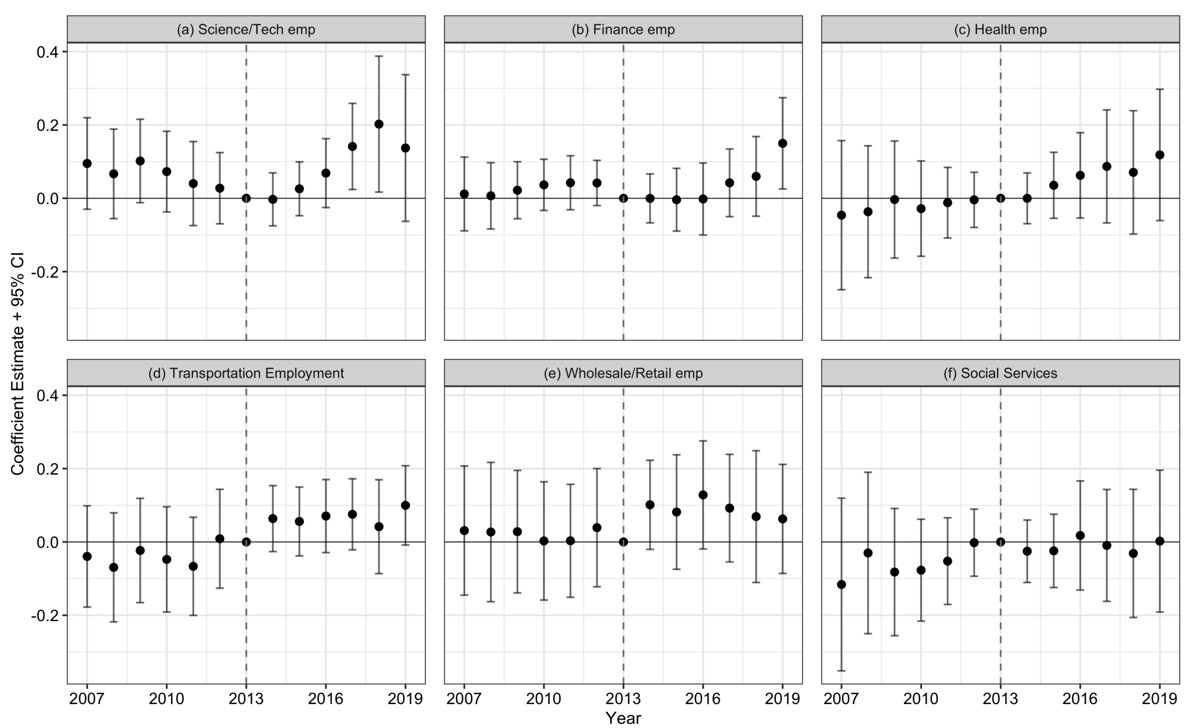
Notes: Event study coefficients and 95% confidence intervals for unrest per 100,000 sector workers. Rows show sectors and columns show incident scale, with sector definitions as in Figure 5. The denominator is secondary-sector employment for manufacturing and construction incidents, and tertiary-sector employment for transport and service incidents. The vertical axis varies across panels.

Figure A5: Per Capita Labor Unrest with Megacities as Control (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for overall per capita labor unrest (events per 100,000 registered population), using megacities as the reference group. Triangles represent treatment cities. Circles represent control cities.

Figure A6: Service Employment by Subsector (2007–2019)



Notes: Event study coefficients and 95% confidence intervals for log employment in six service subsectors. The top row shows (a) scientific and technical services, (b) finance and insurance, and (c) healthcare. The bottom row shows (d) transportation and communication, (e) wholesale and retail trade, and (f) social services.

A2 Additional Tables

Table A1: CLB Incident Counts by Year and City Size

	By City Size						Analysis Sample	
Year	Size 1	Size 2	Size 3	Size 4	Size 5	Total	Treatment	Control
2007	14	19	20	12	4	69	16	39
2008	17	18	38	13	0	86	13	56
2009	4	6	17	5	3	35	8	23
2010	27	20	34	4	5	90	9	54
2011	31	44	86	29	5	195	34	130
2012	82	80	158	45	14	379	59	238
2013	117	135	246	105	20	623	125	381
2014	201	208	555	265	82	1,311	347	763
2015	474	402	1,059	559	157	2,651	716	1,461
2016	485	323	1,038	550	184	2,580	734	1,361
2017	288	147	449	266	75	1,225	341	596
2018	363	191	648	346	105	1,653	451	839
2019	317	191	526	251	61	1,346	312	717

Notes: Total CLB-recorded labor unrest incidents (strikes and protests combined) by year and city-size category. "Treatment" pools sizes 4–5 (population below 1 million). "Control" pools sizes 2–3 (population 1–5 million).

Table A2: Pre-Reform Sample Characteristics

	Treatment (<1M)		Control (1–5M)		Diff.
	Mean	SD	Mean	SD	
Registered population (10,000s)	61.33	(21.00)	169.55	(75.77)	-108.22
Per capita GRP (yuan)	42,250.33	(29,439.69)	51,315.42	(39,749.82)	-9,065.09
Secondary employment	54,759.86	(42,982.12)	186,058.87	(230,164.85)	-131,299.00
Tertiary employment	51,742.31	(24,335.94)	159,869.48	(170,202.26)	-108,127.17
Average wage (yuan)	33,627.18	(11,110.64)	35,606.05	(11,311.66)	-1,978.87
Public revenue per capita	3,703.38	(4,053.32)	4,995.76	(5,599.19)	-1,292.38
Public expenditure per capita	6,607.65	(4,754.46)	6,716.60	(5,789.37)	-108.95
Industrial enterprises	193.78	(191.10)	990.43	(1,515.37)	-796.66
Unrest events (count)	0.26	(0.70)	1.11	(4.34)	-0.85
Unrest events per 100k	0.04	(0.11)	0.06	(0.19)	-0.02
Cities	154		119		

Notes: Pre-reform means (2007–2013) for treatment cities (sizes 4–5, population below 1 million) and control cities (sizes 2–3, population 1–5 million). Population is registered (Hukou) population in units of 10,000. Per capita GRP, public revenue, and public expenditure are in yuan per registered resident. "Diff." reports the difference in means (treatment minus control).

Table A3: Robustness Check, Classification Using 2014 Population

	Unrest (1)	Strike (2)	Protest (3)
Treat × Post	0.122***	0.052***	0.069**
	(0.033)	(0.011)	(0.029)
Observations	3,205	3,205	3,205
R^2	0.64	0.49	0.61

Notes: Treat × Post coefficients for per capita unrest (events per 100,000 registered population) using 2014 population for city size classification instead of the baseline 2013 classification. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A4: Pre-Trends Joint F-Tests

Specification	F-statistic	p-value	Pre-reform coefficients
Unrest/100k	1.00	0.422	6
Strike/100k	0.76	0.603	6
Protest/100k	0.40	0.882	6
Manufacturing Large	1.73	0.109	6
Manufacturing Small	0.64	0.672	5
Registered population	1.54	0.160	6
Secondary employment	1.14	0.334	6
Tertiary employment	0.59	0.739	6

Notes: Joint F-tests that all pre-reform (2007–2012) event study coefficients equal zero, for the headline per capita outcomes, the manufacturing subcategories (Large = 100+ participants, Small = fewer than 100), and the registered-population and sectoral-employment event studies.

Table A5: Count-Based Estimates (PPML)

	Unrest (1)	Strike (2)	Protest (3)
Treat × Post	0.24^{**}	0.44^{***}	0.24
	(0.10)	(0.12)	(0.19)
Observations	3,163	3,002	2,502

Notes: Treat × Post coefficients from PPML regressions using raw event counts. PPML coefficients are semi-elasticities. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A6: Difference-in-Differences Estimates for Log Registered Population

	(Log) Population (1)
Treat × Post	0.058^{*}
	(0.033)
Dep. var. mean (log)	4.496
Observations	3,487
R^2	0.95

Notes: Treat × Post coefficient for log registered population. Post = 1 for years after 2013. Dep. var. mean reports the pre-reform sample mean (2007–2013). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A7: Heterogeneity by Industry and Incident Scale, Per Capita Rates

	Estimate	N
Manufacturing		
Small (1–100 participants)	0.019*	3,549
	(0.012)	
Large (100+ participants)	0.013***	3,549
	(0.004)	
Construction		
Small	0.035**	3,549
	(0.014)	
Large	0.002	3,549
	(0.003)	
Transport		
Small	0.027***	3,549
	(0.005)	
Large	0.005*	3,549
	(0.003)	
Services		
Small	0.001	3,549
	(0.006)	
Large	0.002	3,549
	(0.003)	

Notes: Each row reports the pooled DiD estimate (Treat × Post) for per capita unrest (events per 100,000 registered population) in the indicated sector-scale category. Sector and scale definitions as in Figure 5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A8: Construction Incident Counts by Period, Scale, and Group

	Small (1–100)		Large (100+)	
Year	Control	Treatment	Control	Treatment
2007	0	0	1	0
2008	0	1	2	1
2009	0	0	1	0
2010	0	0	0	0
2011	2	5	6	2
2012	23	3	9	5
2013	3	1	3	3

Notes: CLB-recorded construction-sector incidents by period, incident scale, and treatment group. Small = 1–100 participants. Large = 100+ participants. Construction events are sparse before electronic CLB coverage begins in 2011; small-scale construction appears only once in 2007–2010, in treatment cities in 2008.

Table A9: Heterogeneity by Industry and Scale, Per Sector-Worker Rates

	Estimate	N
Manufacturing		
Small (1–100 participants)	0.398*	3,549
	(0.227)	
Large (100+ participants)	0.187*	3,549
	(0.097)	
Construction		
Small	0.724*	3,549
	(0.377)	
Large	-0.008	3,549
	(0.046)	
Transport		
Small	0.205***	3,549
	(0.061)	
Large	0.052	3,549
	(0.033)	
Services		
Small	-0.014	3,549
	(0.056)	
Large	0.013	3,549
	(0.027)	

Notes: Treat × Post coefficients for unrest per 100,000 sector workers. Manufacturing and construction use secondary-sector employment as the denominator. Transport and services use tertiary-sector employment. Sector and scale definitions as in Figure 5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A10: OLS vs. PPML on Per Capita Outcomes

	OLS			PPML		
	(1)	(2)	(3)	(4)	(5)	(6)
Treat × Post	0.104***	0.042***	0.060**	0.304**	0.506***	0.231
	(0.033)	(0.010)	(0.028)	(0.150)	(0.170)	(0.227)
Observations	3,487	3,487	3,487	3,158	2,997	2,498

Notes: Columns (1)–(3) use OLS; columns (4)–(6) use Poisson pseudo-maximum likelihood (PPML). OLS coefficients are level effects; PPML coefficients are log-multiplicative. $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A11: DiD Estimates with Megacities as Control

	Unrest (1)	Strike (2)	Protest (3)
Panel A. Time-Varying Population Denominator			
1–5M × Post	0.0006	0.024*	-0.023
	(0.091)	(0.014)	(0.090)
<1M × Post	0.103	0.066***	0.035
	(0.092)	(0.016)	(0.089)
Observations	3,643	3,643	3,643
R^2	0.64	0.47	0.62
Panel B. Fixed 2010 Population Denominator			
1–5M × Post	0.042	0.033**	0.009
	(0.098)	(0.015)	(0.093)
<1M × Post	0.177*	0.087***	0.088
	(0.098)	(0.018)	(0.091)
Observations	3,617	3,617	3,617
R^2	0.66	0.49	0.64

Notes: Treat × Post coefficients for per capita unrest (events per 100,000 registered population) using megacities (population above 5 million) as the reference group. "<1M" denotes sizes 4–5. "1–5M" denotes sizes 2–3. Panel A uses time-varying registered population as the denominator. Panel B uses 2010 registered population as a fixed denominator. Post = 1 for years after 2013. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A12: Migrant Composition Response to Hukou Reform (2011–2017)

	Recent (1)	Low Edu. (2)	Intra-prov. (3)	Any Insur. (4)
Treat × Post	0.004	-0.0007	0.007	-0.021**
	(0.008)	(0.005)	(0.006)	(0.008)
Pre-reform control mean	0.386	0.700	0.513	0.301
Individual controls	✓	✓	✓	✓
Observations	485,496	485,496	485,496	196,046
R^2	0.06	0.21	0.26	0.29

Notes: Each column reports the pooled DiD estimate (Treat × Post) for the indicated migrant characteristic. Recent = 1 if arrived within 3 years. Low education = 1 if 9 or fewer years of schooling. Any insurance = 1 if holding any social insurance. Individual controls include gender, marital status, and Hukou type. All specifications include city, province-by-year, and occupation-by-year fixed effects with standard errors clustered at the city level. Data from the CMDS. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A13: Employment Responses by Sector

Panel A. Broad Sectors

	Secondary (1)	Tertiary (2)
Treat × Post	-0.041	0.042*
	(0.045)	(0.022)
Observations	3,476	3,476
R^2	0.95	0.97

Panel B. Service Subsectors

	SciTech (1)	Finance (2)	Health (3)	Transport (4)	Retail (5)	SocialServ (6)
Treat × Post	0.038	0.018	0.081	0.101*	0.071	0.040
	(0.063)	(0.036)	(0.081)	(0.053)	(0.056)	(0.078)
Observations	3,476	3,476	3,194	3,476	3,476	3,190
R^2	0.91	0.92	0.88	0.92	0.90	0.92

Notes: Treat × Post coefficients for log employment. Panel A reports secondary and tertiary sectors. Panel B decomposes tertiary employment into six subsectors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A14: CMDS Migrant Sample Summary Statistics

	Treatment (<1M)		Control (1–5M)	
	Pre	Post	Pre	Post
Log monthly wage	7.854	8.056	7.869	8.150
	(0.553)	(0.575)	(0.507)	(0.525)
Age	35.687	36.763	34.836	36.084
	(8.822)	(9.662)	(8.589)	(9.432)
Male	0.549	0.551	0.522	0.538
	(0.498)	(0.497)	(0.500)	(0.499)
Low education ($\leq$ 9 yrs)	0.753	0.695	0.704	0.650
	(0.431)	(0.461)	(0.456)	(0.477)
Married	0.839	0.848	0.831	0.839
	(0.368)	(0.359)	(0.375)	(0.368)
Any social insurance	0.173	0.147	0.273	0.259
	(0.378)	(0.354)	(0.446)	(0.438)
Recent migrant ($\leq$ 3 yrs)	0.373	0.299	0.386	0.334
	(0.484)	(0.458)	(0.487)	(0.472)
Manufacturing	0.137	0.143	0.246	0.234
	(0.344)	(0.350)	(0.431)	(0.423)
Construction	0.098	0.078	0.082	0.074
	(0.297)	(0.268)	(0.275)	(0.261)
Transport	0.126	0.139	0.121	0.131
	(0.332)	(0.346)	(0.326)	(0.338)
Observations	51,546	97,822	108,148	206,798

Notes: Summary statistics from the China Migrants Dynamic Survey (CMDS), 2011–2017. Treatment cities have population below 1 million (sizes 4–5). Control cities have population 1–5 million (sizes 2–3). "Pre" covers 2011–2013; "Post" covers 2014–2017. Low education is defined as 9 or fewer years of schooling. Recent migrant is defined as having arrived in the current city within the past 3 years. Standard deviations in parentheses.

Table A15: Robustness of CMDS Migrant Results (2011–2017)

	All (1)	Mfg. (2)	Constr. (3)	Transport (4)	Other Svc. (5)
Panel A. Migrant Wage Robustness					
(a) Baseline	-0.037***	-0.017	-0.041***	-0.038**	-0.041***
	(0.009)	(0.012)	(0.015)	(0.015)	(0.012)
(b) No Occ×Year FE	-0.045***	-0.021	-0.037**	-0.039**	-0.051***
	(0.010)	(0.013)	(0.016)	(0.015)	(0.012)
(c) No Hukou control	-0.037***	-0.017	-0.040***	-0.038**	-0.041***
	(0.009)	(0.012)	(0.015)	(0.015)	(0.012)
Pre-reform control mean	7.885	7.876	8.028	7.777	7.892
Observations	424,723	99,096	36,890	59,115	229,622
Panel B. Sector Choice Robustness					
(a) Baseline		0.003	-0.004	-0.001	0.017***
		(0.004)	(0.002)	(0.002)	(0.005)
(b) No Occ×Year FE		0.006	-0.007*	0.004	0.015**
		(0.006)	(0.004)	(0.004)	(0.008)
(c) No Hukou control		0.003	-0.004	-0.001	0.017***
		(0.004)	(0.002)	(0.002)	(0.005)
Pre-reform control mean		0.246	0.082	0.121	0.441
Observations		485,496	485,496	485,496	485,496

Notes: Panel A reports the Treat × Post coefficient with log monthly wage as the dependent variable. Panel B uses a sector indicator as the dependent variable. Row (a) is the baseline from Table 4. Row (b) drops occupation-by-year FE. Row (c) drops the Hukou type control. All specifications include individual controls, city FE, and province-by-year FE. Standard errors clustered at the city level. Data from the CMDS (2011–2017). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A16: Migrant Wage Effects by Tenure (2011–2017)

	Mfg. (1)	Constr. (2)	Transport (3)	Other Svc. (4)
Treat × Post	-0.021	-0.037**	-0.032*	-0.034**
	(0.014)	(0.017)	(0.018)	(0.013)
Treat × Post × Recent	0.017	-0.006	-0.014	-0.020
	(0.019)	(0.022)	(0.021)	(0.013)
Pre-reform control mean	7.876	8.028	7.777	7.892
Observations	99,096	36,890	59,115	229,622
R^2	0.36	0.28	0.26	0.24

Notes: Each column reports the Treat × Post coefficient and its interaction with a Recent indicator (arrived within 3 years). Columns restrict to migrants in the indicated sector. Individual controls, city, province-by-year, and occupation-by-year fixed effects. Standard errors clustered at the city level. Data from the CMDS (2011–2017). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A17: Economic and Fiscal Responses

	Average Wage (1)	Revenue p.c. (2)	Expenditure p.c. (3)
Treat × Post	-0.006	-0.074*	-0.052
	(0.014)	(0.039)	(0.035)
Observations	3,476	3,487	3,487
R^2	0.95	0.94	0.90

Notes: Treat × Post coefficients for log outcomes. Columns report (1) log average wages of employed persons, (2) log local public revenue per capita, and (3) log local public expenditure per capita. All specifications include city and province-by-year fixed effects with standard errors clustered at the city level. Megacities are excluded. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.